Further Appendices for "When does IV identification not restrict outcomes?"

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G Illustrative examples from the brute force search

G.1 Binary treatment and binary instrument

With a binary treatment and binary instrument, the brute-force search reveals that there are exactly two selection models that admit of outcome-nonrestrictive identification of treatment effects. The first is the classic LATE model of Imbens and Angrist (1994) (Example 1), and the second is Example 2 from the main text.

Consider first Example 1. Row reduction of the matrices $A^{[1]}$ and $A^{[0]}$ given in Example 1 in Section 3 yields:

$$rs(A^{[1]}) = span \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \quad \text{and} \quad rs(A^{[0]}) = span \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

This leads to the two planes depicted in Figure 1.

Remark: In the binary-binary LATE model, the rank k of $A^{[t]}$ is $k = |\mathcal{Z}| = 2$ for either $t = 0$ or $t = 1$, and in either case $|rs(A^{[t]}) \cap \{0,1\}^n| = 2$. This does not meet the upper bound of $2^k = 4$ from Melo and Winter (2019) (see Footnote 8). However the result does imply that there can be no more than $2^{|\mathcal{Z}|}$ binary combinations, even though typically $2^{|\mathcal{Z}|} < 2^{|\mathcal{G}|}$ and there are $2^{|\mathcal{G}|}$ potential values of c to consider ex-ante.

In the case of example 2,[1] the matrix A becomes:

	compliers	defiers
$\mathbf{z = 0}$	0	1
$\mathbf{z = 1}$	1	0

The rowspaces of $A^{[1]}$ and $A^{[0]}$ are the same and both span $\mathbb{R}^2$: $rs(A^{[1]}) = rs(A^{[0]}) = span \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$. Thus, given the results of Section 3, we know that treatment effect parameters that are outcome-nonrestrictive identified correspond to any non-zero vertex of the unit cube in $\mathbb{R}^2$, as depicted in Figure 1 below. Note that by Theorem 1, $\mathbb{E}[D_i|Z_i = 1]$ and $1 - \mathbb{E}[D_i|Z_i = 0]$ are both measures for the same

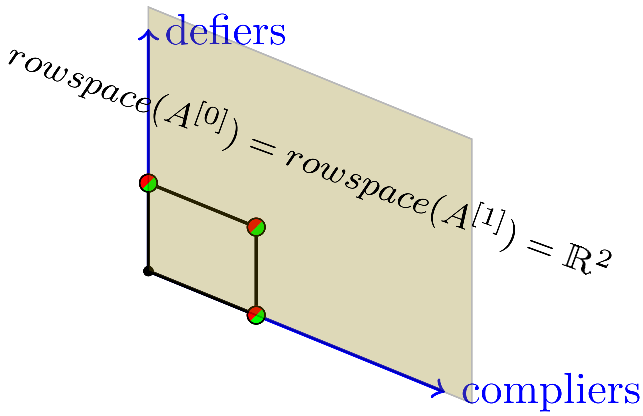


Figure 1: A geometric depiction of the model with compliers and defiers only. The vectors $c = (0,1)'$, $c = (1,0)'$ and $c = (1,1)'$ all belong to both $rs(A^{[1]})$ and $rs(A^{[0]})$ and hence the LATE for either response type or the ATE are identified. As in Figure 1, the split-shading of a given vertex (red/green in color) of the unit square indicates that it lies in $rs(A^{[0]}) \cap rs(A^{[1]})$ and is not equal to the zero vector.

population parameter $P(c(G_i) = 1) = P(i \text{ is complier})$. Thus $\mathbb{E}[D_i|Z_i = 1] = 1 - \mathbb{E}[D_i|Z_i = 0]$ can be used as an overidentification restriction for this selection model (there is no such restriction for the LATE model that simply rules out defiers).

[1]Goff and Lee (2024) apply this selection model to study the effect of an NFL team deferring the kickoff on the game outcome, with the kickoff coin flip that decides which team is given the option to defer as the instrument. If it is common knowledge between the teams whether receiving the kickoff is beneficial in that particular game, then a simple model of optimizing play would predict that each game will either be a "complier" or a "defier".

G.2 3 treatments, binary instrument

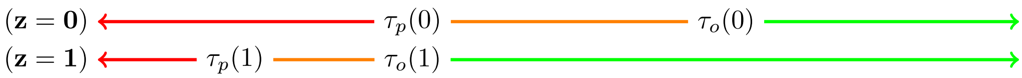
Now suppose that the instrument is binary and $\mathcal{T} = \{0, 1, 2\}$. With no restrictions on selection behavior, there are $3^{|\mathcal{Z}|} = 9$ conceivable response types. Table 1 reports that in this case there are five selection models that afford a total of five distinct outcome-nonrestrictive identification results. These results are all listed in Appendix K.

As an example that may be empirically relevant, consider a selection model that has $T(z)$ increasing in z, but rules out always-1 takers and individuals that skip from $t = 0$ to $t = 2$ when z is increased from 0 to 1. This restriction leaves four response types: always-0 takers, always-2 takers, individuals who move from treatment 0 to treatment 1, and individuals who move from treatment 1 to treatment 2. SM.3.2.1 reported in Appendix K reveals that two treatment effects $\Delta_c^{t,t'}$ are identified in this selection model. For example, the quantity

$$\frac{\mathbb{E}[Y_i \cdot D_i^{[1]}|Z_i = 1]}{\mathbb{E}[D_i^{[1]}|Z_i = 1]} - \frac{\mathbb{E}[Y_i \cdot D_i^{[0]}|Z_i = 0] - \mathbb{E}[Y_i \cdot D_i^{[0]}|Z_i = 1]}{\mathbb{E}[D_i^{[0]}|Z_i = 0] - \mathbb{E}[D_i^{[0]}|Z_i = 1]}$$

corresponds to the binary collection with $\alpha_0 = (0, 1)'$ and $\alpha_1 = (1, -1)'$, and identifies $\mathbb{E}[Y_i(1) - Y_i(0)|T_i(0) = 0, T_i(1) = 1]$. The treatment effect $\mathbb{E}[Y_i(2) - Y_i(1)|T_i(0) = 1, T_i(1) = 2]$ is identified by a similar estimand.

This selection model could represent a setting in which $t = 2$ represents passing a test outright, $t = 1$ passing the test "provisionally", and $t = 0$ failing. Suppose that a reform implemented for some schools lowers the score threshold τ_o for an outright pass to the old threshold τ_p for a provisional pass, while further lowering τ_p, as depicted below:



Students will then belong to one of the four types described above, depending on their test score. The quantity $\mathbb{E}[Y_i(1) - Y_i(0)|T_i(z) = z]$ represents the average effect of moving from a fail to a provisional pass among the students who are brought in to a provisional pass by the grading reform.

The selection model of Kline and Walters (2016): Another observation from the $|\mathcal{T}| = 3, |\mathcal{Z}| = 2$ setting is that the selection model of Kline and Walters (2016) (KW) does not appear in the catalog of Appendix K. KW study a setting in which the binary instrument is an offer to choose $t = 2$, while treatments $t = 0$ and $t = 1$ are always available even if $z = 0$. On the grounds of revealed preference, KW impose that $T_i(1) \neq T_i(0) \implies T_i(1) = 2$ which results in a selection model with five response types. KW show that the parameter $\mathbb{E}[Y_i(2) - Y_i(T_i(0))|T_i(1) = 2, T_i(0) \neq 2]$ is then identified. The quantity $T_i(0)$ represents an individual's next preferred alternative to $t = 2$, which may vary across those i for whom $T_i(1) = 2, T_i(0) \neq 2$. As a result, this parameter does not fit the form of the general family of treatment effect parameters $\Delta_c^{t,t'}$ introduced in Section A. Indeed, the brute force search confirms that unfortunately *no* parameters of the form $\Delta_c^{t,t'}$ between two fixed treatmets t and t' are identified in the KW selection model.

G.3 Binary treatment, 3 instrument values

Suppose now that treatment is binary and $\mathcal{Z} = \{0, 1, 2\}$. Table 1 reports that in this case there are 11 selection models that afford a total of 30 distinct outcome-nonrestrictive identification results, listed in Appendix K. One observation that emerges when we extend the analysis to instruments that take more than two values is that, there now exist binary collections that require coefficients α_z that do not belong to the set $\{-1, 0, 1\}$. Although this is entirely consistent with Proposition 3 for $|\mathcal{Z}| \geq 3$, a reasonable conjecture ex-ante might have been that $\alpha_z \in \{-1, 0, 1\}$ always holds, given the preponderance of this pattern in known identification results (see for example all of the results surveyed in Appendix H).

For the sake of exposition, let us consider a "judge-IV" setting in which defendants i receive a bail decision from a randomly-assigned judge z, with $t = 1$ indicating that the defendent remains incarcerated and $t = 0$ that they are released on bail. Suppose that the defendants are one of three types $g \in \mathcal{G}$ (typically unobserved to the researcher), comprising the columns of the table below. "Prepared" defendants dress formally and speak politely in their bail hearing, perhaps also presenting evidence that they are not a danger to the community. "Unprepared" defendants do not make such efforts. A third category of "flight-risk" defendants are thought to be particularly capable of and likely to fail to appear for trial if they are granted bail (while this is not true of the first two groups, e.g. due to strong personal ties to the jurisdiction or insufficient financial means to leave town).

	prepared	unprepared	flight-risk
$\mathbf{z = 0}$ (standard)	0	0	1
$\mathbf{z = 1}$ (character)	0	1	0
$\mathbf{z = 2}$ (skeptics)	1	0	1

The above table summarizes selection behavior when the judges also belong to one of three types, represented across rows. "Standard" judges are only concerned with failure to appear, and keep only the flight-risk defendants incarcerated. "Character" judges instead attempt to infer the risk of a defendant to public safety on the basis of the defendant's presentation and arguments to their character made in the bail hearing, but do not attempt to assess whether the defendant is likely to skip town. "Skeptic" judges are also sensitive to judgments about presentation, but in the opposite direction: they are suspicious of defendants precisely when they seem to be making a case that they are not dangerous. They deny bail for the prepared defendants, and also deny bail for flight-risk defendants.[2]

Note that this model does not satisfy the strong LATE monotonicity assumption typically invoked in judge-IV settings, which has been challenged on empirical grounds (Frandsen et al., 2023; Sigstad, 2023). If there were no skeptic ($z = 2$) judges, then this model would instead consist of compliers, defiers, and never-takers, which we have already seen permits no outcome-nonrestrictive identification results for treatment effects. However, the presence of the skeptics aids here in identification, as we can then identify the average effect of incarceration among two groups $\mathbb{E}[Y_i(1) - Y_i(0)|G_i \in \{unprepared, flight\text{-}risk\}]$ by $\frac{\mathbb{E}[Y_iD_i|Z_i=0]+\mathbb{E}[Y_iD_i|Z_i=1]}{\mathbb{E}[D_i|Z_i=0]+\mathbb{E}[D_i|Z_i=1]} - \frac{-\mathbb{E}[Y_iD_i|Z_i=0]+\mathbb{E}[Y_iD_i|Z_i=1]+2\cdot\mathbb{E}[Y_iD_i|Z_i=2]}{-\mathbb{E}[D_i|Z_i=0]+\mathbb{E}[D_i|Z_i=1]+2\cdot\mathbb{E}[D_i|Z_i=2]}$, corresponding to the binary collection with $\alpha_1 = (1, 1, 0)$ and $\alpha_0 = (-1, 1, 2)'$.[3] Note that using this result requires judge types to be observable, or estimable from judges each seeing many cases.

G.4 3 treatment values, 3 instrument values

In the case of $\mathcal{Z} = \mathcal{T} = \{0, 1, 2\}$, the brute force approach returns 251 distinct binary collections spread across 251 unique selection models. These results nest for example two identification results presented in Kirkeboen, Leuven and Mogstad (2016) (KLM). KLM consider an unordered treatment which represents a student's field of study, where students are "assigned" to a given field, i.e $Z_i = j$ represents an incentive to choose field j. Proposition 2 of KLM presents three special cases in which a two stage least squares estimand with indicators for treatments 1 and 2 instrumented by indicators for $Z_i = 1$ and $Z_i = 2$

[2] This selection model is equivalent to SM.2.3.4 in Appendix K, after permuting instrument/treatment labels. Note that this model is merely illustrative: more types and nuance in their definitions could add some realism.

[3] A coefficient of two is inevitable for $t = 0$, since we need $\alpha_1 = 1$ (using the α_z notation) for $c(flightrisk) = 0$, $\alpha_0 = -\alpha_1$ to get $c(prepared) = 0$, but can only achieve $c(unprepared) = 1$ if $\alpha_2 + \alpha_1 = 1$.

recovers causally interpretable coefficients. While their first result (restricting treatment effects to be homogeneous) is not outcome-nonrestrictive, the other two results are.

For example, the second result in KLM Proposition 2 shows that if preferences are further restricted so that $D_i^{[2]}(1) = D_i^{[2]}(0)$ and $D_i^{[1]}(2) = D_i^{[1]}(0)$ for all i (an offer to one program does not affect whether or not the student chooses the other program), then $\mathbb{E}[Y_i(1) - Y_i(0)|D_i^{[1]}(1) > D_i^{[1]}(0)]$ and $\mathbb{E}[Y_i(2) - Y_i(0)|D_i^{[2]}(2) > D_i^{[2]}(0)]$ are each identified.[4] Let us consider how the first of these results appears in the comprehensive search (the second result proceeds similarly). Upon a relabeling of treatment/instrument values and removing one response type,[5] selection model SM.3.3.63 in the catalog amounts to the following: $A = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 1 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 2 & 2 & 1 & 2 & 1 \end{bmatrix}$. This selection model has seven response types, whereas the selection model considered by KLM contains only the first six columns of A. In the larger selection model with all seven groups, the treatment effect $\mathbb{E}[Y_i(1) - Y_i(0)|T_i(1) \neq T_i(0)]$ is identified by the binary collection with $\alpha_0 = (-2, 1, 1)'$ and $\alpha_1 = (1, -1, 0)'$.[6] Thus, we have seen that KLM's selection model can be relaxed to allow an additional response type, with the same estimand that identifies $\mathbb{E}[Y_i(1)-Y_i(0)|D_i^{[1]}(1) > D_i^{[1]}(0)]$ in their more restrictive model identifying $\mathbb{E}[Y_i(1)-Y_i(0)|T_i(1) \neq T_i(0)]$ more generally. Code available from the author allows one to check in general whether a given selection model can be relaxed in this way, using the catalog of identification results (available for $|\mathcal{T}|, |\mathcal{Z}| \leq 3$).

Another example that we can analyze given the 251 identification results revealed is the model of Cheng and Small (2006). Cheng and Small (2006) consider a setting where $\mathcal{Z}, \mathcal{T} = \{0, A, B\}$ with groups summarized by the matrix: $A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ A & A & 0 & 0 \\ B & 0 & B & 0 \end{bmatrix}$ with the first, second, and third rows corresponding to instrument values 0,A, and B, respectively. This selection model corresponds to what Cheng and Small (2006) call Monotonicity I. They give an application in which it is plausible to further assume that the column 00B does not occur, and they refer to this more restrictive selection model as Monotonicity II. Even the larger Monotonicity I model is nested within many of the larger selection models revealed by the comprehensive search. For example, starting with SM.3.3.22, we can associate the labels "1" "2" "3" with "A","0', and "B", respectively. Then associating the instrument labels "1" "2" "3" with "0","B', and "A' respectively, SM.3.3.22 becomes $A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 0 & 0 & 2 & 2 & 2 & 2 & 2 \\ 0 & 2 & 0 & 2 & 1 & 0 & 0 & 2 & 2 \end{bmatrix}$. The catalog shows that the parameter $\Delta_c^{1,2}$ is identified with $c = (1, 1, 0, 0, 1, 0, 0, 0, 0)'$. This implies, in the notation of Cheng and Small (2006), that the treatment effect parameter $\mathbb{E}[Y_i(B) - Y_i(A)|G_i = 0A0 \text{ or } G_i = 0AB]$ is identified. However, enumerating through all such selection models, one could confirm the statement by Cheng and Small (2006) that the parameters $\mathbb{E}[Y_i(A) - Y_i(0)|G_i = 0A0]$ and $\mathbb{E}[Y_i(A) - Y_i(0)|G_i = 0AB]$ are *not* point identified (even when imposing Monotonicity II), without restricting outcomes.

G.5 4 treatment values, 4 instrument values: spillover effects within pairs

Although the $|\mathcal{T}| = |\mathcal{Z}| = 4$ case is not included in the brute-force search of Table 1 (due to the computational burden), it remains easy to check for outcome-nonrestrictive identification in any given

[4] Throughout, KLM also maintain a version of unordered monotonicity (cf. Heckman and Pinto 2018) in which $D_i^{[1]}(1) \geq D_i^{[1]}(0)$ and $D_i^{[2]}(2) \geq D_i^{[2]}(0)$: an offer of admission never causes a student to select out of that field.

[5] In particular, label the treatments $(0, 1, 2)$ as $(1, 0, 2)$, swap instrument values 1 and 2, and drop column 6. All results for the 3×3 case are enumerated in a working paper version of this paper: `https://arxiv.org/abs/2406.02835`.

[6] Accordingly, $\mathbb{E}[Y_i(1) - Y_i(0)|T_i(1) \neq T_i(0)] = \frac{\mathbb{E}[Y_i \cdot D_i^{[1]}|Z_i=2]+\mathbb{E}[Y_i \cdot D_i^{[1]}|Z_i=1]-2\cdot\mathbb{E}[Y_i \cdot D_i^{[1]}|Z_i=0]}{\mathbb{E}[D_i^{[1]}|Z_i=2]+\mathbb{E}[D_i^{[1]}|Z_i=1]-2\cdot\mathbb{E}[D_i^{[1]}|Z_i=0]} - \frac{\mathbb{E}[Y_i \cdot D_i^{[0]}|Z_i=1]-\mathbb{E}[Y_i \cdot D_i^{[0]}|Z_i=0]}{\mathbb{E}[D_i^{[0]}|Z_i=1]-\mathbb{E}[D_i^{[0]}|Z_i=0]}$. Identification of $\mathbb{E}[Y_i(2) - Y_i(0)|T_i(2) \neq T_i(0)]$ is analogous.

selection model using the results of Section 3. This section presents an alternative application in the 4×4 case to supplement the study of interaction effects from Section 5.

Consider a setting in which each unit i has one "neighbor" $n(i)$, and we allow for violations of SUTVA within neighbor pairs $(i, n(i))$. This can be accommodated by expanding the set of treatments $\mathcal{T}$ to accommodate values of such pairs, so that $Y_i = Y_i(T_i, T_{n(i)})$ where $T_{n(i)}$ is the treatment of the neighbor of unit i, indexed by $n(i)$. I consider the case in which treatment T itself is binary, so that $\tilde{T}_i := (T_i, T_{n(i)})$ may take one of four values $(0,0), (1,0), (0,1), (1,1)$. Following the notation in Section 5, we denote these pair-level "treatments" as $0, A, B, C$, where $\tilde{T}_i = 0$ indicates that neither unit is treated, $\tilde{T}_i = A$ that only unit i is treated, $\tilde{T}_i = B$ that only their neighbor is treated, and $\tilde{T}_i = C$ that both i and their neighbor is treated.

For each i, let Z_i be a binary instrument that reflects whether i is "assigned" to receive treatment. See Kang and Imbens (2016) and Vazquez-Bare (2023) for related setups. Let $\tilde{T}_i(z, z')$ reflect the treatments for the pair as a function of treatment assignments (z, z') for the pair. Let $\tilde{Z}_i := (Z_i, Z_{n(i)})$ be the pairs realized treatment assignment, which can take any of four counterfactual values $z \in \tilde{\mathcal{Z}} = \{0, A, B, C\}$. I maintain throughout two assumptions about selection: i) first, that the T_i component of $\tilde{T}_i(z, z')$ only depends on z, and that the $T_{n(i)}$ component of $\tilde{T}_i(z, z')$ only depends on z'; and ii) secondly, that each selection uptake is monotonic such that $T_i(1) \geq T_i(0)$, where we write $T_i(z, z')$ as $(T_i(z), Y_{n(i)}(z'))$.

These restrictions leave nine response types, enumerated in the table below:

assigned ↓	(nt,nt)	(cm,cm)	(cm,nt)	(nt,cm)	(at,at)	(at,cm)	(cm,at)	(nt,at)	(at,nt)
0=(0,0)	0	0	0	0	C	A	B	B	A
A=(1,0)	0	A	A	0	C	A	C	B	A
B=(0,1)	0	B	0	B	C	C	B	B	A
C=(1,1)	0	C	A	B	C	C	C	B	A

Given a function $c(\cdot)$, the local average direct effect of one's own treatment on their outcome is:

$$\begin{aligned} LADTE &:= \mathbb{E}[Y_i(1, Z_{n(i)}) - Y_i(0, Z_{n(i)}) | c(G_i) = 1] \\ &= P(Z_{n(i)} = 1) \cdot \mathbb{E}[Y_i(C) - Y_i(B) | c(G_i) = 1] + P(Z_{n(i)} = 0) \cdot \mathbb{E}[Y_i(A) - Y_i(0) | c(G_i) = 1] \end{aligned}$$

using that $P(Z_{n(i)} = 1 | c(G_i) = 1) = P(Z_{n(i)} = 1)$ by independence.

Similarly, the local average spillover (indirect) effect is

$$\begin{aligned} LASTE &:= \mathbb{E}[Y_i(Z_i, 1)) - Y_i(Z_i, 0) | c(G_i) = 1] \\ &= P(Z_i = 1) \cdot \mathbb{E}[Y_i(C) - Y_i(A) | c(G_i) = 1] + P(Z_i = 0) \cdot \mathbb{E}[Y_i(B) - Y_i(0) | c(G_i) = 1] \end{aligned}$$

Since the distributions of Z_i and $Z_{n(i)}$ are identified, we can then point identify the LADTE and the LASTE provided that we can identify μ_t^c for all of $t \in \{0, A, B, C\}$. An application of Theorems 1 and 2 shows that this is possible without restricting outcomes in the above selection model if and only if $c(g) = \mathbb{1}(g = cm, cm)$.

This can be shown by direct enumeration of the 2^9 possible functions $c : \mathcal{G} \to \{0, 1\}$. The vector forms α_t of the resulting coefficient functions $\alpha^{[t]}(z)$ are:

$$\alpha_0 = (+1, -1, -1, +1)', \quad \alpha_A = (-1, +1, +1, -1)'$$
$$\alpha_B = (-1, +1, +1, -1)', \quad \alpha_C = (+1, -1, -1, +1)'$$

Similar to the case of complementarities, the selection model therefore implies the overidentification restriction that

$$\begin{aligned}
p &:= P(T_i=0|Z_i=(1,1)) - P(T_i=0|Z_i=(0,1)) - P(T_i=0|Z_i=(1,0)) + P(T_i=0|Z_i=(0,0)) \\
&= -P(T_i=A|Z_i=(1,1)) + P(T_i=A|Z_i=(0,1)) + P(T_i=A|Z_i=(1,0)) - P(T_i=A|Z_i=(0,0)) \\
&= -P(T_i=B|Z_i=(1,1)) + P(T_i=B|Z_i=(0,1)) + P(T_i=B|Z_i=(1,0)) - P(T_i=B|Z_i=(0,0)) \\
&= P(T_i=C|Z_i=(1,1)) - P(T_i=C|Z_i=(0,1)) - P(T_i=C|Z_i=(1,0)) + P(T_i=C|Z_i=(0,0))
\end{aligned}$$

for some value $p \in [0,1]$, which identifies $P(g = \text{cm,cm})$.

H Recovering existing identification results as binary combinations and collections

The notions of binary combinations and binary collections reveals a common structure among several existing IV point identification results.

H.1 Example 1: LATE monotonicity and marginal treatment effects

Here I extend my analysis of the monotonicity assumption of Imbens and Angrist (1994) to cases with more than two instrument values. Treatment remains binary $\mathcal{T} = \{0,1\}$. Since $D_i^{[0]}(z) = 1 - D_i^{[1]}(z)$, we can focus on the single treatment indicator $D_i(z) := D_i^{[1]}(z)$. Imbens and Angrist (1994) assume that:

Assumption IAM. *For all $z, z' \in \mathcal{Z}$: $D_i(z) \geq D_i(z')$ for all i or $D_i(z) \leq D_i(z')$ all i.*

Suppose z, z' are a pair such that the former case of assumption IAM obtains, and define a binary combination with $K = 2$, $z_1 = z'$, $\alpha_1 = 1$, $z_2 = z$, $\alpha_2 = -1$. Then Eq. (4) from the main paper yileds

$$\mathbb{E}[Y_i(1)|D_i(z') > D_i(z)] = \frac{\mathbb{E}\left[Y_i \cdot D_i|Z_i = z'\right] - \mathbb{E}\left[Y_i \cdot D_i|Z_i = z\right]}{\mathbb{E}\left[D_i|Z_i = z'\right] - \mathbb{E}\left[D_i|Z_i = z\right]} \tag{1}$$

and similarly

$$\mathbb{E}[Y_i(0)|D_i(z') > D_i(z)] = \frac{\mathbb{E}\left[Y_i \cdot (1-D_i)|Z_i = z'\right] - \mathbb{E}\left[Y_i \cdot (1-D_i)|Z_i = z\right]}{\mathbb{E}\left[(1-D_i)|Z_i = z'\right] - \mathbb{E}\left[(1-D_i)|Z_i = z\right]}$$

Combining, we have that $\mathbb{E}[Y_i(1) - Y_i(0)|D_i(z') > D_i(z)] = \frac{\mathbb{E}[Y_i|Z_i=z'] - \mathbb{E}[Y_i|Z_i=z]}{\mathbb{E}[D_i|Z_i=z'] - \mathbb{E}[D_i|Z_i=z]}$, which is Theorem 1 of Imbens and Angrist (1994).

Suppose that $\mathcal{Z}$ is continuous and for all $u \in [0,1]$ there exists a $z \in \mathcal{Z}$ such that $P(z) := P(D_i = 1|Z_i = z) = u$. Let $U_i = \inf_{z \in \mathcal{Z}}\{P(z) : D_i(z) = 1\}$. Given IAM, U_i plays the role of G_i, indicating the "first" instrument value (as ordered by the propensity score function $P(z)$) at which i would take treatment. For any given u, let z be a point in $\mathcal{Z}$ such that $P(z) = u$ and take a sequence z_j in $\mathcal{Z}$ such that $z_j \to z$ as $j \to \infty$. Taking the limit of Eq. (1) we have that:

$$\mathbb{E}[Y_i(1)|U_i = u] = \lim_{j\to\infty} \frac{\mathbb{E}\left[Y_i \cdot D_i|Z_i = z_j\right] - \mathbb{E}\left[Y_i \cdot D_i|Z_i = z\right]}{\mathbb{E}\left[D_i|Z_i = z_j\right] - \mathbb{E}\left[D_i|Z_i = z\right]} = \frac{d}{du}\mathbb{E}\left[Y_i \cdot D_i|P(Z_i) = u\right]$$

and similarly for $Y_i(0)$, allowing us to identify marginal treatment effects (cf. Heckman et al. (2006)).

H.2 Example 2: Vector monotonicity (Goff 2024)

Goff (2024) considers a binary treatment and finite $\mathcal{Z} \subseteq \mathcal{Z}_1 \times \mathcal{Z}_2 \times \dots \mathcal{Z}_J$, and the following monotonicity assumption.

Assumption 2 (vector monotonicity). *There exists an ordering $\geq_j$ on $\mathcal{Z}_j$ for each $j \in \{1 \dots J\}$ such that for all $z, z' \in \mathcal{Z}$, if $z \geq z'$ component-wise according to the $\{\geq_j\}$, then $D_i(z) \geq D_i(z')$ for all i.*

Theorem 1 of Goff (2024) shows that average counterfactual means are identified under vector monotonicity for groups defined by the condition $c(G_i, Z_i) = 1$, where c satisfies a condition called "Property M" and Z_i has full rectangular support. His Proposition 6 shows that Property M is equivalent to $c(G_i, Z_i) = \sum_{k=1}^{K} \alpha_k \cdot D_i^{[t]}(z_k(Z_i))$ where K is an even number no greater than $J/2$ and $\alpha_k = (-1)^k$ with $z_{k+1}(z) \geq z_k(z)$ component-wise according to the orders $\geq_j$, for all k. In what follows I for simplicity focus on the special case of target parameters in which c depends on G_i only, and not additionally on Z_i. See Appendix I for a discussion of other parameters.

In the case of two binary instruments $J = 2$, there are six selection types compatible with vector monotonicity, with names introduced by Mogstad et al. 2021:

	Z_1 comp.	Z_2 comp.	eager-comp.	reluctant-comp.	n.t.	a.t.
$\mathbf{z} = (\mathbf{0}, \mathbf{0})'$	0	0	0	0	0	1
$\mathbf{z} = (\mathbf{1}, \mathbf{0})'$	1	0	1	0	0	1
$\mathbf{z} = (\mathbf{0}, \mathbf{1})'$	0	1	1	0	0	1
$\mathbf{z} = (\mathbf{1}, \mathbf{1})'$	1	1	1	1	0	1

With this table defining matrix $A^{[1]}$, some algebra shows that for example $c = (1, 0, 0, 1, 0, 0)'$ occurs in the rowspace of both $A^{[1]}$ and $A^{[0]}$. One way to see this is to work out the row reduced echelon forms of $A^{[1]}$ and $A^{[0]}$, which preserve their row-spaces and are:

$$rref(A^{[1]}) = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \qquad rref(A^{[0]}) = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

From the first row of each of the reduced echelon forms, we can see immediately that e.g. the average treatment effect among Z_1 and reluctant compliers is outcome-nonrestrictive identified. Adding all four rows we see that the average treatment effect among all of the four compliers types is outcome-nonrestrictive identified, what Goff (2024) calls the "all-compliers LATE". Goff (2024) shows how these and similar point identification results generalize to any number of instruments under vector monotonicity.

H.3 Example 3: Unordered Monotonicity, Heckman and Pinto (2018)

Heckman and Pinto (2018) (HP) consider a finite $\mathcal{Z}$ and assume what they call *unordered monotonicity* (UM) for a multi-valued treatment:

Assumption UM. *For any $t \in \mathcal{T}$ and $z, z' \in \mathcal{Z}$, either $D_i^{[t]}(z) \geq D_i^{[t]}(z')$ for all i or $D_i^{[t]}(z') \geq D_i^{[t]}(z)$ for all i.*

Given UM, let us fix a $t \in \mathcal{T}$ and label the points in $\mathcal{Z}$ as z_m for $m = 1, 2 \dots |\mathcal{Z}|$, where the points are labeled in increasing order of the propensity score for treatment t: $P_t(z_{m+1}) \geq P_t(Z_m)$, where we define $P_t(z) = \mathbb{E}[D_i^{[t]}|Z_i = z]$. The order is not important in the case of ties. Let $\Sigma_{ti} = |\{z \in \mathcal{Z} : T_i(z) = t\}|$ be the number of $z \in \mathcal{Z}$ for which i takes treatment t. Note that Σ_{ti} is exactly equal to $|\mathcal{Z}| - m + 1$ for the smallest m such that $D_i^{[t]}(z_m) = 1$. Thus we have a binary combination for any treatment t and value $s \in \{0, 1, \dots |\mathcal{Z}|\}$: in particular $D_i^{[t]}(z_m) - D_i^{[t]}(z_{m-1}) \in \{0, 1\}$ for all i, and is equal to i for those units having $\Sigma_{ti} = s$.

It then follows immediately from Eq. (4), as in the IAM case with a binary treatment (cf. Eq. 1), that $E[Y_i(t)|\Sigma_{ti} = s]$ is identified for any $s = 1 \dots |\mathcal{Z}|$ as:

$$\mathbb{E}[Y_i(t)|\Sigma_{ti} = s] = \begin{cases} \frac{\mathbb{E}\left[Y_i \cdot D_i^{[t]}|Z_i = z_m\right] - \mathbb{E}\left[Y_i \cdot D_i^{[t]}|Z_i = z_{m-1}\right]}{\mathbb{E}\left[D_i^{[t]}|Z_i = z_m\right] - \mathbb{E}\left[D_i^{[t]}|Z_i = z_{m-1}\right]} & \text{if } s < |\mathcal{Z}| \\ \frac{\mathbb{E}\left[Y_i \cdot D_i^{[t]}|Z_i = z_1\right]}{\mathbb{E}\left[D_i^{[t]}|Z_i = z_1\right]} & \text{if } s = |\mathcal{Z}| \end{cases} \tag{2}$$

where $m = |\mathcal{Z}| - s + 1$.

This provides a simple proof of HP's Theorem T-6, which shows that $E[Y_i(t)|\Sigma_{ti} = s]$ is identified. HP show that

$$E[Y_i(t)|\Sigma_{ti} = s] = \frac{c' {A^{[t]}}^+ Q_Z}{c' {A^{[t]}}^+ P_Z} \tag{3}$$

where B^+ is the Moore-Penrose pseudo-inverse of a matrix B, Q_Z is a vector of $\mathbb{E}[Y_i D_i^{[t]}|Z_i = z]$ across z and P_Z is a vector of $\mathbb{E}[D_i^{[t]}|Z_i = z]$ across z. Here c corresponds to our parameter of interest (indexed by the pair (t, s)), with an entry of one if $\Sigma_{tg} = s$ for that selection type and zero otherwise.

To see the equivalence between this result and (2), we can take advantage of the structure of $A^{[t]}$ under UM to replace it with a smaller matrix whose inverse is very simple. Note that any two selection types g sharing a value of Σ_{ti} will have identical entries in c, and will have identical corresponding columns in the matrix $A^{[t]}$. This implies that they will have identical rows in ${A^{[t]}}^+$. We can remove the redundant columns of $A^{[t]}$ by indexing columns by values of Σ_{ti} rather than by full response vectors g, and similarly indexing elements of c by values of Σ_{ti}. This yields the same vector $c'A^{[t]+}$ as before, up to a scalar factor that counts the number of values of g such that $\Sigma_{ti} = s$. However, this factor cancels out in the numerator and denominator of (3). With this modification, $A^{[t]}$ is now a $|\mathcal{Z}|$ by $|\mathcal{Z}|+1$ matrix and c is now a standard basis vector equal to one in its s^{th} element (and zero elsewhere).

Let us now order the rows of this modified $A^{[t]}$ according to z_1, z_2, etc, and it's columns in decreasing order of Σ_{ti}. With this ordering, $A^{[t]}$ is simply a lower triangular matrix of ones, appended to the right by a single column of zeros. It can then be verified that rows $s = 2\dots(|\mathcal{Z}|-1)$ of $A^{[t]+}$ are of the form $(0,\dots,-1,1,\dots 0)'$ with $s-2$ zeroes on the left (while the first row is composed of a single 1 in the first column, and the last row is all zeros).[7] Note that given the definition of c, $c'A^{[t]+}$picks out the s^{th} row of $A^{[t]+}$ in (3), and we have that (2) and (3) are equivalent.

Remark 7.1 of HP observes that given the above, treatment effects can be identified if (in my notation) for some s, s' and $t, t' \in \mathcal{T}$, $\mathbb{1}(\Sigma_{ti} = s) = \mathbb{1}(\Sigma_{t'i} = s')$ almost surely, since then we can identifiy $\mathbb{E}[Y_i(t') - Y_i(t)|\Sigma_{t'i} = s'] = \mathbb{E}[Y_i(t')|\Sigma_{t'i} = s'] - \mathbb{E}[Y_i(t)|\Sigma_{ti} = s]$. The idea of binary collections can be though of as a generalization of this type of result beyond the case of unordered monotonicity.

H.4 Example 4: Lee and Salanié (2018)

Lee and Salanié (2018) (LS) consider a class of models in which unit i's selection type depends upon a J-dimensional vector $V_i \in [0,1]^J$ and a vector valued function $Q : \mathcal{Z} \to \mathcal{Q}$ where $\mathcal{Q} \subseteq \mathbb{R}^J$. Selection is assumed to follow:

$$D_i^{[t]}(z) = \sum_{l \subseteq \{1\dots J\}} c_l^t \cdot \prod_{j \in l} \mathbb{1}(V_{ji} \leq Q_j(z)) \tag{4}$$

for some set of coefficients c_l^t defined over the subsets of $\{1\dots J\}$, for each $t \in \mathcal{T}$, and where V_{ji} is the j^{th} component of V_i, and Q_j the j^{th} component of Q. This model nests the marginal treatment effects (MTE) framework when we have a binary treatment and $J = 1$, in which case we may let $Q(z) = \mathbb{E}[D_i|Z_i = z]$ be the propensity score function.

The second part of LS's Theorem 3.1 shows that under support/regularity conditions:

$$E[Y_i(t)|V_i = q] = \frac{\frac{\partial^J}{\partial_{q_1}\dots\partial_{q_J}}\mathbb{E}[Y_i D_i^{[t]}|Q(Z_i) = q]}{\frac{\partial^J}{\partial_{q_1}\dots\partial_{q_J}}\mathbb{E}[D_i^{[t]}|Q(Z_i) = q]} \tag{5}$$

Now let's see how this result can also be obtained through Theorem 1. For any vector $q \in \mathbb{R}^J$, let $S_i(q) := \{j \in \{1\dots J\} : V_{ji} \leq q_j\}$ be the set of indices for which $V_{ji} \leq q_j$.

[7] E.g. the modified forms with $|\mathcal{Z}| = 4$ are $A^{[t]} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix}$, $A^{[t]+} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$.

Then $D_i^{[t]}(z) = \sum_{l\subseteq S_i(Q(z))} c_l^t$. Note that $D_i^{[t]}(z)$ only depends on z through $Q(z)$. Thus, we could for each q consider an arbitrary value $z \in \mathcal{Z}$ such that $Q(z) = q$, call it $Q^{-1}(q)$, and think of $D_i^{[t]}$ as a function $D_i^{[t]}(Q^{-1}(q))$ of q.

Let us consider a binary combination constructed to capture all units such that V_i belongs to a rectangle $(q, q+h_1]\times(q, q+h_2]\cdots\times(q, q+h_J]$ in $\mathbb{R}^J$ for some "corner" location $q \in \mathbb{R}^J$ and widths $h_1 \dots h_J$. For any $s \subseteq \{1\dots J\}$, let $h_s := \sum_{j\in s} h_j\mathbf{e}_j$, where $\mathbf{e}_j$ is the j^{th} standard basis vector. This takes the form of a binary combination (α, t) having $K = 2^J$ and coefficients $\alpha_k = (-1)^{|s_k|}/\lambda$ for a certain scalar λ. The corresponding instrument values are $z_k = Q^{-1}(q + h_s)$ given an arbitrary ordering $s_1 \dots s_K$ on the K distinct subsets of $\{1\dots J\}$. Below we will verify that the corresponding linear combination of the $D_i^{[t]}(Q^{-1}(q))$ is equal to $c(G_i)$ with probability one, where we let $c(G_i)$ be an indicator for $V_i \in (q, q+h_1]\cdots\times(q, q+h_J]$. Via Eq. (4) in the main text, we can thus identify $\mathbb{E}[Y_i(t)|V_i \in (q, q+h_1]\cdots\times(q, q+h_J]]$ as:

$$\mathbb{E}[Y_i(t)|c(G_i) = 1] = \frac{\sum_{s\subseteq\{1\dots J\}}(-1)^{|s|}\cdot\mathbb{E}\left[Y_i\cdot D_i^{[t]}|Z_i = Q^{-1}(q+h_s)\right]}{\sum_{s\subseteq\{1\dots J\}}(-1)^{|s|}\cdot\mathbb{E}\left[D_i^{[t]}|Z_i = Q^{-1}(q+h_s)\right]} \tag{6}$$

The scalar λ depends on the selection mechanism (4) and is $\lambda := \sum_{s\subseteq\{1\dots J\}}(-1)^{|s|}\sum_{l\subseteq s} c_l^t$.[8]

We now verify that with this notation $c(G_i) = \sum_{k=1}^{2^J}\alpha_k\cdot D_i^{[t]}(z_k)$. That is:

$$c(G_i) = \frac{1}{\lambda}\sum_{s\subseteq\{1\dots J\}}(-1)^{|s|}\cdot D_i^{[t]}\left(Q^{-1}(q+\sum_{j\in s}h_j\mathbf{e}_j)\right) = \frac{1}{\lambda}\sum_{s\subseteq\{1\dots J\}}(-1)^{|s|}\sum_{l\subseteq S_i(q+h_S)} c_l^t \tag{7}$$

Note that for any $S' \subseteq S$, $S_i(q+h'_S) \subseteq S_i(q+h_S)$. Thus $S_i(q+h_{\{1\dots J\}})$ is the "largest" $S_i(q+h_s)$ and $S_i(q)$ is the smallest. Define:

$$A_i = S_i(q+h_{\{1\dots J\}}) - S_i(q) = \{j \in \{1\dots J\} : q < V_{ji} \leq q + h_{\{1\dots J\}}\}$$

A_i is simply the set of dimensions in which V_{ji} falls within the rectangle starting at q with widths $h_{\{1\dots J\}}$. Now comes the crucial step: we'll now show that (7) is zero for any individual i for which A_i does not contain all of $\{1\dots J\}$. Indeed, if there were any $j \notin A_i$, each set S in the first summation of (7) that did not contain j would be canceled out by the set $S\cup j$, because $(-1)^{|S\cup j|} = -(-1)^{|S|}$, while $S_i(q+h_S) = S_i(q+h_{S\cup j})$. Pairing all sets in this way, we see that evaluates to zero unless $A_i = \{1\dots J\}$. Now,

[8] We can simplify this expression of λ as follows. Note that given 4) the coefficients c_l^t must be such that $\sum_{l\subseteq s} c_l^t \in \{0,1\}$ for any $s \subseteq \{1\dots J\}$. Let S_t be the collection of s such that it is equal to one. This is the collection of subsets of the thresholds that when crossed correspond to taking treatment t. Then $\lambda = \sum_{s\in S_t}(-1)^{|s|}$. We can derive an alternative expression for λ by making use of the identity that for any $\sum_{f\subseteq S}(-1)^{|f|} = 0$ for any $S \neq \emptyset$. Then:

$$\lambda = \sum_{l\subseteq\{1\dots J\}} c_l^t \sum_{s\supseteq l}(-1)^{|s|} = \sum_{l\subseteq\{1\dots J\}} c_l^t\left(\cancel{\sum_{s\subseteq\{1\dots J\}}(-1)^{|s|}} - \cancel{\sum_{s\subseteq l}(-1)^{|s|}} + (-1)^{|l|}\right) = \sum_{l\subseteq\{1\dots J\}}(-1)^{|l|}\cdot c_l^t$$

$A_i = \{1 \dots J\}$ implies that $S_i(q) = \emptyset$, and we can now write $c(G_i)$ as:

$$c(G_i) = \mathbb{1}(A_i = \{1 \dots J\}) \cdot \frac{1}{\lambda} \cdot \left(\sum_{s \subseteq \{1 \dots J\}} (-1)^{|s|} \sum_{l \subseteq s} c_l^t \right) = \mathbb{1}(A_i = \{1 \dots J\})$$

observing that we've defined c to be equal to the quantity in parentheses, which depends on the selection model but not on i.

LS's Theorem 3.1 considers the J^{th} order derivative

$$\begin{aligned}
&\frac{\partial^J}{\partial_{q_1} \dots \partial_{q_J}} \mathbb{E}[Y_i D_i^{[t]} | Q(Z_i) = q] \\
&\qquad = \frac{\partial^J}{\partial_{q_2} \dots \partial_{q_J}} \lim_{h_1 \downarrow 0} \frac{1}{h_1} \left(\mathbb{E}[Y_i D_i^{[t]} | Q(Z_i) = q + h_1 \mathbf{e}_1] - \mathbb{E}[Y_i D_i^{[t]} | Q(Z_i) = q] \right) \\
&\qquad = \lim_{h_1 \dots h_J \downarrow 0} \frac{1}{\prod_{j=1}^J h_j} \cdot \sum_{s \subseteq \{1 \dots J\}} (-1)^{|s|} \cdot \mathbb{E}\left[Y_i D_i^{[t]} \middle| Q(Z_i) = q + \sum_{j \in s} h_j \mathbf{e}_j \right]
\end{aligned}$$

and takes the ratio:

$$\frac{\frac{\partial^J}{\partial_{q_1} \dots \partial_{q_J}} \mathbb{E}[Y_i D_i^{[t]} | Q(Z_i) = q]}{\frac{\partial^J}{\partial_{q_1} \dots \partial_{q_J}} \mathbb{E}[D_i^{[t]} | Q(Z_i) = q]} = \lim_{h_1 \dots h_J \downarrow 0} \frac{\sum_{s \subseteq \{1 \dots J\}} (-1)^{|s|} \cdot \mathbb{E}\left[Y_i D_i^{[t]} \middle| Q(Z_i) = q + \sum_{j \in s} h_j \mathbf{e}_j \right]}{\sum_{s \subseteq \{1 \dots J\}} (-1)^{|s|} \cdot \mathbb{E}\left[D_i^{[t]} \middle| Q(Z_i) = q + \sum_{j \in s} h_j \mathbf{e}_j \right]}$$

LS's result (5) thus considers the limit of Eq. (6) as the width of the rectangle goes to zero in all dimensions.

H.5 Example 5: Unordered (generalized) partial monotonicity

We can define a generalization of vector and partial monotonicity to settings with multi-valued treatments, also nesting unordered monotonicity:

Assumption UPM. *For any $t \in \mathcal{T}$, there exists a partial order $\succeq_t$ on $\mathcal{Z}$, such that if $z' \succeq_t z$, $D_i^{[t]}(z') \geq D_i^{[t]}(z)$ for all i.*

Note that even in the case of a binary treatment, UPM represents a generalization of partial monotonicity (PM), defined by Mogstad et al. (2019) for settings with multiple instruments. UPM allows for an arbitrary partial order on $\mathcal{Z}$, while PM considers a partial order that is based on holding all instruments but one at fixed values.

Assumption UPM implies that for any such z, z': $D_i^{[t]}(z') - D_i^{[t]}(z) \in \{0, 1\}$ and thus

$$\mathbb{E}[Y_i(t) | D_i^{[t]}(z') > D_i^{[t]}(z)] = \frac{\mathbb{E}[Y_i D_i^{[t]} | Z_i = z'] - \mathbb{E}[Y_i D_i^{[t]} | Z_i = z]}{\mathbb{E}[D_i^{[t]} | Z_i = z'] - \mathbb{E}[D_i^{[t]} | Z_i = z]}$$

UPM holds, for example, when instruments correspond to choice sets and agents choose rationally from them, as in Arora et al. (2021). In such a setting instrument values z are subsets of the treatments $\mathcal{T}$ that are available to the agent, and $D_i^{[t]}(z) \geq D_i^{[t]}(z')$ whenever ($z/t \subseteq z'/t$ and $t \in z$ if $t \in z'$). In words, $D_i^{[t]}(z)$ is weakly increasing with respect to the inclusion of t in z (since i can only choose t if it is available), and weakly decreasing with respect to the inclusion of any $t' \neq t$ in z (since i may prefer t' to t).

H.6 Example 6: Pairwise notions of monotonicity

Sun and Wüthrich (2024) proposes a notion of IV-validity that is specific to two values z, z' of the instrument (which may be a vector). This includes the standard LATE model assumptions (independence, exclusion, and monotonicity). However, if independence and exclusion are maintained, the notion of pairwise valid instruments reduces to what we might call *pairwise-monotonicity*, i.e. that $D_i^{[t]}(z') \geq D_i^{[t]}(z')$ almost surely, or vice versa.

van't Hoff, Lewbel and Mellace (2025) consider a notion of "limited monotonicity" for settings with multiple binary instruments and a binary treatment, which in the notation above corresponds to a setting in which $z' = (1, \dots 1)$ and $z = (0, \dots 0)$. van't Hoff (2023) extends this notion to ordered treatments that need not be binary.

In the context of "judge designs" where the instrument is a scalar continuous measure of "leniency" with respect to a binary treatment, Sigstad (2023) and Sigstad (2024) introduce a notion of "extreme-pair" monotonicity $D_i(\bar{j}) \geq D_i(\underline{j})$ almost surely, where $\bar{j}$ is the strictest judge, and $\underline{j}$ the most lenient.

In the case of a binary treatment D_i, the above papers point out that under a limited version of "monotonicity" between a pair of values z, z', a particular local average treatment effect can be identified from a simple Wald estimand:

$$\mathbb{E}[Y_i(1) - Y_i(0)|D_i(z') > D_i(z)] = \frac{\mathbb{E}[Y_i|Z_i = z'] - \mathbb{E}[Y_i|Z_i = z]}{\mathbb{E}[D_i|Z_i = z'] - \mathbb{E}[D_i|Z_i = z]}$$

This corresponds to a binary collection in which $\alpha_{z'} = 1$ and $\alpha_z = -1$ for $t = 1$, while $\alpha_{z'} = -1$ and $\alpha_z = 1$ for $t = 0$.

I Letting local causal parameters depend on Z_i

Let $z_k : \mathcal{Z} \to \mathcal{Z}$ be a function that maps an instrument value Z_i to some possibly different value in $\mathcal{Z}$. Non-constant functions $z_k(\cdot)$ will allow us to nest parameters such as the average treatment effect on the treated, as well as some parameters from Goff (2024). In that paper $z_k(z)$ could for instance change one component of z, and the α_k and $z_k(\cdot)$ can be chosen so that $c(G_i, Z_i) := \sum_k \alpha_k \cdot D_i^{[t]}(z_k(Z_i))$ only takes values of zero or one, i.e. $\alpha_z = \{\alpha_k, z_k(z)\}_{k=1}^K$ yields a binary combination for any $z \in \mathcal{Z}$. Then, by the law of iterated expectations: $\mathbb{E}\left[Y_i(t)\,|c(G_i, Z_i) = 1\right] = \sum_{z \in \mathcal{Z}} P(Z_i = z) \cdot \mathbb{E}[Y_i(t)|c(G_i, Z_i) = 1]$ where each term in the summand is identified by (4) and the distribution of Z_i.

Let us maintain the assumption that the support of the instruments $\mathcal{Z}$ is discrete and finite. Consider any counterfactual mean of the form $\theta = \mathbb{E}[Y_i(t)|c(G_i, Z_i) = 1]$ where now $c : \mathcal{G} \times \mathcal{Z} \to \{0, 1\}$. By the law of iterated expectations over Z_i and independence

Eq. (2), we can write θ as:

$$\theta = \sum_{z \in \mathcal{Z}} P(Z_i = z | c(G_i, z) = 1) \cdot \mathbb{E}[Y_i(t) | c(G_i, z) = 1, Z_i = z]$$
$$= \sum_{z \in \mathcal{Z}} P(Z_i = z) \cdot \mathbb{E}[Y_i(t) | c_z(G_i) = 1] \tag{8}$$

where we let $c_z(g)$ denote $c(g, z)$. Eq (8) shows that θ can be written as a convex combination of $|\mathcal{Z}|$ counterfactual means of the form μ_c^t considered by Theorems 1 and 2, with complier groups $c_z(\cdot)$ that depend on z. It is clear then by Theorem 1, a sufficient condition for θ to be outcome-nonrestrictive identified is that c_z lies in the rowspace of matrix $A^{[t]}$ for each $z \in \mathcal{Z}$.

Theorem 2 similarly extends to the more general class of functions $c(G_i, Z_i)$, provided that the family $\mathscr{P}_Z$ of distributions over the instruments allows for degenerate distributions at each value of Z_i. Then c_z must lie in the rowspace of $A^{[t]}$ for all $z \in \mathcal{Z}$. If it were not, then for some $z \in \mathcal{Z}$, $c_z \notin rs(A^{[t]})$ and hence $\mu_{c_z}^t$ is not outcome-nonrestrictive identified, by Theorem 2. For a degenerate distribution $P_{\mathcal{Z}}$ that sets $P(Z_i = z) = 1$, $\theta = \mu_{c_z}^t$ and hence θ is not outcome-nonrestrictive identified if $c_z \notin rs(A^{[t]})$. With this extension Theorem 2 of this paper nests Theorem 2 of Goff (2024) as a special case, and expands its reach even in the case that vector monotonicity is maintained, if the outcome variable is continuous.

J Partial identification when $c \notin rowspace(A^{[t]})$

J.1 Relationship to Bai, Huang, Moon, Shaikh and Vytlacil (2024)

Bai et al. (2024) (BHMSV) study the identifying power for ATEs and unconditional counterfactual means of a restriction on selection that they call *generalized monotonicity* (GM). In my notation, GM says that for a given $\mathcal{P}_{latent}$ and each $t \in \mathcal{T}$, there exists an instrument value $z^* = z^*(t, \mathcal{P}_{latent})$ such that

$$P(D_i(z^*) \neq t \text{ and } D_i(z) = t \text{ for some } z \in \mathcal{Z}) = 0 \tag{9}$$

according to $\mathcal{P}_{latent}$. That is, no individual takes treatment t when $z \neq z^*$ unless they also do when $z = z^*$. BHMSV show that GM or any strengthening of it (that does not restrict outcomes) does not reduce the size of identified sets for unconditional parameters of the form $\mathbb{E}[Y_i(t)]$, when the outcome variable has finite support $\mathcal{Y}$ and the instruments are also finite.

While GM nests many notions of monotonicity from the literature that have been used for positive point identification results, it generalizes them in a different way than the criterion $c \in rs(A^{[t]})$ of the present paper does. While $c \in rs(A^{[t]})$ ensures point identification of $\mathbb{E}[Y_i(t)|c_{G_i} = 1]$, GM represents a double-edged sword when the parameter of interest is an unconditional mean or ATE with $c = (1, 1, \ldots 1)'$. Using Theorem 2 of this paper, we can see that GM is in fact sufficient to establish either that i) $\mathbb{E}[Y_i(t)]$ is point identified in an outcome-nonrestrictive way; or ii) that it is *not* point identified in an outcome-nonrestrictive way. Which of these cases i) or ii) holds can be determined by the observable distribution $\mathcal{P}_{obs}$, and does not depend on $\mathcal{G}$ beyond it satisfying GM.

Let $\tilde{\mathcal{G}}(\mathcal{P}_{latent})$ be the support of G_i under $\mathcal{P}_{latent}$, and note that (9) is equivalent to:

$$\text{For all } g \in \tilde{\mathcal{G}}(\mathcal{P}_{latent}): \quad A^{[t]}_{z^*,g} = 0 \implies A^{[t]}_{z,g} = 0 \text{ for all } z \in \mathcal{Z} \tag{10}$$

Consider a given distribution of observables $\mathcal{P}_{obs}$. Either $P(T_i = t|Z_i = z^*) = 1$ or $P(T_i = t|Z_i = z^*) < 1$ according to $\mathcal{P}_{obs}$. If the first case holds, then $\mathbb{E}[Y_i(t)] = \mathbb{E}[Y_i|Z_i = z^*]$ and $\mathbb{E}[Y_i(t)]$ is thus point-identified without requiring any restrictions on selection. Thus, assuming GM or any strengthening of it cannot reduce the identified set for $\mathbb{E}[Y_i(t)]$ further, unless it results in model rejection.

If on the other hand $P(T_i = t|Z_i = z^*) < 1$ and GM holds, then there must exist a $g \in \tilde{\mathcal{G}}(\mathcal{P}_{latent})$ such that $A^{[t]}_{z^*,g} = 0$. Therefore by (10), for this g it must be that $A^{[t]}_{z,g} = 0$ for all $z \in \mathcal{Z}$, i.e. there are never-takers with respect to treatment t. This in turn implies that $(1, 1, \dots 1)' \notin rs(A^{[t]})$, precisely the case in which we know that $\mathbb{E}[Y_i(t)]$ is *not* outcome-nonrestrictive point identified, by Theorem 2.

By showing that the bounds on $\mathbb{E}[Y_i(t)]$ in partially identified settings are not improved by imposing restrictions stronger than GM, BHMSV underscore the importance of: i) focusing on other parameters of interest beyond the ATE (i.e. $c \neq (1, 1, \dots 1)'$) when one is willing to impose restrictions on selection; and ii) finding restrictions on selection that are outside of the scope of GM. Indeed, many of the selection models reported in Appendix K below do not satisfy GM, yet are sufficient for point identification of more localized treatment effect parameters than the ATE (and in some cases the ATE as well).

The remainder of this section provides more detail to build intuition about the connection between BHMSV's result and the proof of Theorem 2 in this paper. For a given $\mathcal{P}_{obs}$, let us write the identified set for $\mathbb{E}[Y_i(t)]$ under model M as

$$\Theta(\mathcal{P}_{obs}, M) = \{\theta(\mathcal{P}) : \phi(\mathcal{P}) = \mathcal{P}_{obs} \text{ and } \mathcal{P} \in M\}$$

Given BHMSV's assumption that $\mathcal{Y}$ is finite, let us for each $y \in \mathcal{Y}$ define x^y to be a $|\mathcal{G}|$-component vector with components $x^y_g = P(Y_i(t) = y|G_i = g)$ and β to be a $|\mathcal{Z}|$-component vector with components $\beta^y_z = P(Y_i = y, T_i = t|Z_i = z)$. The restriction $\phi(\mathcal{P}) = \mathcal{P}_{obs}$ corresponds to the set of solutions to finite system of linear equations $A^{[t]}x^y = \beta^y$, for each $y \in \mathcal{Y}$. Given $|\mathcal{Y}| < \infty$, we can collect these into a single finite linear system $\mathcal{A}^{[t]}\tilde{x} = \tilde{\beta}$, where $\mathcal{A}^{[t]}$ is a block diagonal matrix of $A^{[t]}$ copied $|\mathcal{Y}|$ times, $\tilde{x}$ is a $|\mathcal{Y}| \times |\mathcal{G}|$ component vector, and $\tilde{\beta}$ is a $|\mathcal{Y}| \times |\mathcal{Z}|$ component vector. The set $\mathcal{X} := \{\tilde{x}(\mathcal{P}) : \phi(\mathcal{P}) = \mathcal{P}_{obs}\}$ is thus a vector space, where we let $\tilde{x}(\mathcal{P})$ represent $\tilde{x}$ as a function of the distribution of model fundamentals $\mathcal{P}$.

Whether GM or any strengthening of it reduces the identified set for $\mathbb{E}[Y_i(t)]$ thus depends upon whether the action of $\theta(\cdot)$ on the $\mathcal{P} \in M$ such that $\tilde{x}(\mathcal{P}) \in \mathcal{X}$ reduces $\Theta(\mathcal{P}_{obs}, M)$ relative to a case with no restrictions on selection. Eq. (22) from the proof of Theorem 2 suggests that $\Theta(\mathcal{P}_{obs}, M)$ satisfies

$$\Theta(\mathcal{P}_{obs}, M) \subseteq \left\{ \mathbb{1}'(A^{[t]})^{+}\beta + \sum_{g'} [\mathbb{1}'(I - (A^{[t]})^{+}A^{[t]})]_{g'} \cdot w_{g'} : w \in \mathbb{R}^{|\mathcal{G}|} \right\} \tag{11}$$

where $\mathbb{1} := (1, 1, \ldots 1)'$ and β a $|\mathcal{Z}|$-component vector with components $\beta_z = \mathbb{E}[Y_i \cdot \mathbb{1}(T_i = t)|Z_i = z]$. GM implies that the set of the RGS is not a singleton if $P(T_i = t|Z_i = z^*) < 1$. The subset relation appearing in (11) reflects that, as in Theorem 2, some $\tilde{x}$ for which $\mathcal{A}^{[t]}\tilde{x} = \tilde{\beta}$ may not be attainable from $\mathcal{P}$ that are valid distributions and reflect any further assumptions of the model M, for example that Y_i has bounded support.

BHMSV show that if M does not restrict outcomes, $\Theta(\mathcal{P}_{obs}, M)$ is in fact equal to the identified set under no selection restrictions, which is (given the finite support $\mathcal{Y}$):

$$\{\beta_{z^*} - \min\{\mathcal{Y}\} \cdot P(T_i \neq t|Z_i = z^*), \beta_{z^*} + \max\{\mathcal{Y}\} \cdot P(T_i \neq t|Z_i = z^*)\}$$

An interesting question for further study is in what manner the result of BHMSV extends to the more general class target parameters indexed by vectors c that may differ from $(1, 1, \ldots, 1)'$. A reasonable conjecture would be that if, given $\mathcal{P}_{obs}$, a class of restrictions on selection cannot change the fact that $c \notin rs(A^{[t]})$, there is limited scope for such restrictions to reduce the size of the identified set for μ_c^t.

J.2 Partial identification in general

Accordingly, consider an arbitrary $c \in \{0, 1\}^{|\mathcal{G}}$ where we may have that $c \notin rs(A^{[t]})$. By similar logic as above, we can deduce that the identified set $\Theta(\mathcal{P}_{obs}, M)$ for μ_c^t satisfies:

$$\Theta(\mathcal{P}_{obs}, M) \subseteq \frac{1}{P(c(G_i) = 1)} \cdot \left\{ c'(A^{[t]})^{+}\beta + \sum_{g'} [c'(I - (A^{[t]})^{+}A^{[t]})]_{g'} \cdot w_{g'} : w \in \mathbb{R}^{|\mathcal{G}|} \right\}$$

The RHS may again be an outer set for $\Theta(\mathcal{P}_{obs}, M)$, for example when Y_i has bounded support. An added complication now, as compared to unconditional means, is that the probability $P(c(G_i) = 1)$ is no longer known to be equal to one, and our only identifying information for it is that $\sum_{g \in \mathcal{G}} A_{gz}^{[t]} = d_z$ for all $z \in \mathcal{Z}$, where $d_g := P(T_i = t|Z_i = z)$.

K Catalog of outcome-nonrestrictive identification results

The following results are for various small values of $|\mathcal{Z}|$ $|\mathcal{T}|$. Results for $|\mathcal{Z}| = |\mathcal{T}| = 3$ are available upon request from the author (these add roughly 40 pages of output).

For a given $\mathcal{T}$ and $\mathcal{Z}$, binary collections are organized by selection models, given unique identifiers of the format `SM.`$|\mathcal{T}|$`.`$|\mathcal{Z}|$`.s`, where `s` is an index of the various selection models in that setting. Within each selection model, binary collections are enumerated by ascending numbers $i), ii)$ etc. Each binary collection is presented via the coefficient vectors $\alpha_{t'}$ and α_t (following the notation of Sec. 4.2 but keeping t and t' explicit).

Binary collections that share a common maximal selection model $\mathcal{G}(\alpha)$ organized under that selection model, and are not re-listed for $\mathcal{G} \subseteq \mathcal{G}(\alpha)$. Further, some binary collections for $\mathcal{G}$ might be listed under a $\mathcal{G}(\alpha)$ that nests $\mathcal{G}$ only after suitable re-labeling of the treatments and instruments. It is for this reason that the set of binary collections listed under a given selection model may not be closed under addition, even when adding the c for two such collections results in another vector composed of all zeroes and ones.

For example, consider the A matrices for SM.2.3.8 and SM2.3.1 below:

$$\texttt{SM.2.3.8}: \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad\qquad \texttt{SM.2.3.1.swapped}: \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

where by `SM.2.3.1.swapped` I indicate that I have swapped the first and third rows of the A matrix listed in the catalog that follows for SM.2.3.1. This swapping corresponds to a re-labelling of the instrument values.

SM.2.3.8 consists of two selection types, and the catalog shows that it admits of binary collection i) with $\alpha_1 = (0.5, 0.5, 0)'$ and $\alpha_0 = (0, 0, 1)'$ yielding $c = (1, 0)'$ as well as binary collection ii) with $\alpha_1 = (0, 0, 1)'$ and $\alpha_0 = (0.5, 0.5, 0)'$ yielding $c = (1, 0)'$. This implies that SM.2.3.8 also admits of a binary collection yielding $c = (1, 1)'$, with $\alpha_0 = \alpha_1 = (0.5, 0.5, 1)'$.

The reason that this third binary collection is not listed under SM.2.3.8 is that SM.2.3.8 is not maximal for it: unlike collections i) and ii) which just include one of the two types in SM.2.3.8, identification of the average treatment effect for both of the types in SM.2.3.8 holds in the less restrictive selection model SM.2.3.1.swapped, which contains the selection types of SM.2.3.8 in its first and third columns. The sole binary collection listed under SM.2.3.1 corresponds to $c = (1, 1)'$ in SM.2.3.8.

K.1 2 treatments, 2 instrument values

SM.2.2.1

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1)'; \alpha_t = (1, -1)'$; $c = (0, 1, 0)'$

SM.2.2.2

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1)'; \alpha_t = (1, 0)'$; $c = (0, 1)'$

ii) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0)'; \alpha_t = (0, 1)'$; $c = (1, 0)'$

iii) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1)'; \alpha_t = (1, 1)'$; $c = (1, 1)'$

K.2 3 treatments, 2 instrument values

SM.3.2.1

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 \\ 0 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1)'; \alpha_t = (1, -1)'$; $c = (0, 1, 0, 0)'$

SM.3.2.2

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1)'; \alpha_t = (1, -1)'$; $c = (0, 1, 0, 0)'$

SM.3.2.3

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1)'; \alpha_t = (1, 0)'$; $c = (0, 0, 1, 0, 0)'$

SM.3.2.4

$$A = \begin{bmatrix} 2 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1)'; \alpha_t = (1, 0)'$; $c = (0, 1, 0, 0)'$

SM.3.2.5

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1)'; \alpha_t = (1, 1)'$; $c = (1, 1, 0)'$

K.3 2 treatments, 3 instrument values

SM.2.3.1

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'; \alpha_t = (-1, 1, 2)'$; $c = (1, 0, 1)'$

SM.2.3.2

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -2)'$; $\alpha_t = (-1, -1, 2)'$; $c = (0, 1, 1, 0)'$

SM.2.3.3

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, 0.5)'$; $\alpha_t = (1, 1, 1)'$; $c = (1, 1, 1)'$

ii) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 1, 0)'$

iii) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (1, 0, 1)'$; $c = (1, 0, 1)'$

iv) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, -0.5)'$; $\alpha_t = (0, 0, 1)'$; $c = (1, 0, 0)'$

v) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (1, 1, 0)'$; $c = (0, 1, 1)'$

vi) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, -0.5, 0.5)'$; $\alpha_t = (0, 1, 0)'$; $c = (0, 1, 0)'$

vii) $(t', t) = (1, 0)$; $\alpha_{t'} = (-0.5, 0.5, 0.5)'$; $\alpha_t = (1, 0, 0)'$; $c = (0, 0, 1)'$

SM.2.3.4

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (2, 1, -1)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 1, 0)'$

SM.2.3.5

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 1)'$

ii) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'$; $\alpha_t = (-1, 1, 1)'$; $c = (0, 1, 0, 1)'$

iii) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (0, 0, 1)'$; $c = (1, 1, 0, 0)'$

iv) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (0, 1, 0)'$; $c = (0, 0, 1, 1)'$

v) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'$; $\alpha_t = (1, 0, 0)'$; $c = (1, 0, 1, 0)'$

SM.2.3.6

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (2, -1, -1)'$; $\alpha_t = (-2, 1, 1)'$; $c = (0, 1, 1, 0)'$

SM.2.3.7

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 0)'$; $\alpha_t = (0, 0, 1)'$; $c = (1, 0, 0)'$

ii) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, -1, 1)'$; $\alpha_t = (1, 0, 0)'$; $c = (0, 1, 0)'$

SM.2.3.8

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, 0)'$; $\alpha_t = (0, 0, 1)'$; $c = (1, 0)'$

ii) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (0.5, 0.5, 0)'$; $c = (0, 1)'$

SM.2.3.9

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'$; $\alpha_t = (0, 0, 1)'$; $c = (1, 1, 0)'$

ii) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 1)'$; $\alpha_t = (0.5, 0.5, 0.5)'$; $c = (1, 1, 1)'$

iii) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'$; $\alpha_t = (-0.5, 0.5, 0.5)'$; $c = (1, 0, 0)'$

iv) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (0.5, -0.5, 0.5)'$; $c = (0, 1, 0)'$

v) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 1)'$; $\alpha_t = (0, 1, 0)'$; $c = (1, 0, 1)'$

vi) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'$; $\alpha_t = (1, 0, 0)'$; $c = (0, 1, 1)'$

vii) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (0.5, 0.5, -0.5)'$; $c = (0, 0, 1)'$

SM.2.3.10

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'$; $\alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0, 1, 0, 0)'$

SM.2.3.11

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0)'$

ii) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (-1, 1, 0)'$; $c = (0, 0, 1)'$

K.4 3 treatments, 3 instrument values

SM.3.3.1

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -2)'$; $\alpha_t = (-1, 1, 2)'$; $c = (1, 0, 0, 1, 0)'$

SM.3.3.2

$$A = \begin{bmatrix} 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'$; $\alpha_t = (-1, 1, 2)'$; $c = (1, 0, 0, 0, 1, 0)'$

SM.3.3.3

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (2, 1, -1)'$; $\alpha_t = (-1, 1, 2)'$; $c = (1, 1, 0, 0, 0)'$

SM.3.3.4

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'$; $\alpha_t = (-1, 1, 2)'$; $c = (1, 0, 1, 0, 0, 1, 0)'$

SM.3.3.5

$$A = \begin{bmatrix} 0 & 2 & 1 & 0 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-2, 1, 1)'$; $\alpha_t = (-1, 1, 2)'$; $c = (1, 1, 0, 0, 0)'$

SM.3.3.6

$$A = \begin{bmatrix} 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'$; $\alpha_t = (-1, 1, 2)'$; $c = (1, 1, 0, 0, 0, 0)'$

SM.3.3.7

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'$; $\alpha_t = (-1, 1, 2)'$; $c = (1, 1, 1, 0, 0, 0)'$

SM.3.3.8

$$A = \begin{bmatrix} 0 & 2 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 1)'$; $\alpha_t = (-1, 1, 2)'$; $c = (1, 1, 0, 1, 0)'$

SM.3.3.9

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 2)'$; $\alpha_t = (-1, 1, 2)'$; $c = (1, 1, 0, 0, 0)'$

SM.3.3.10

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -2)'$; $\alpha_t = (-1, -1, 2)'$; $c = (0, 1, 1, 0, 0)'$

SM.3.3.11

$$A = \begin{bmatrix} 0 & 1 & 0 & 2 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'; \alpha_t = (-1, -1, 2)'$; $c = (0, 1, 1, 0, 0, 0)'$

SM.3.3.12

$$A = \begin{bmatrix} 0 & 1 & 0 & 2 & 2 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'; \alpha_t = (-1, -1, 2)'$; $c = (0, 1, 1, 0, 0)'$

SM.3.3.13

$$A = \begin{bmatrix} 2 & 1 & 1 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -2)'; \alpha_t = (1, 1, 1)'$; $c = (1, 1, 0, 1, 1, 0)'$

SM.3.3.14

$$A = \begin{bmatrix} 2 & 1 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'; \alpha_t = (1, 1, 1)'$; $c = (1, 1, 0, 0, 1, 1, 0)'$

SM.3.3.15

$$A = \begin{bmatrix} 2 & 1 & 2 & 0 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (2, 1, -1)'; \alpha_t = (1, 1, 1)'$; $c = (1, 1, 0, 1, 0)'$

SM.3.3.16

$$A = \begin{bmatrix} 2 & 1 & 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'; \alpha_t = (1, 1, 1)'$; $c = (1, 1, 1, 1, 0, 1, 1, 0)'$

SM.3.3.17

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, 0.5)'$; $\alpha_t = (1, 1, 1)'$; $c = (1, 1, 1, 0)'$

SM.3.3.18

$$A = \begin{bmatrix} 2 & 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 1)'$; $\alpha_t = (1, 1, 1)'$; $c = (1, 1, 1, 1, 1, 1, 0)'$

SM.3.3.19

$$A = \begin{bmatrix} 0 & 2 & 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & 2 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -2)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 0, 1, 0, 0)'$

SM.3.3.20

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, -1)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 0, 1, 0, 0, 0)'$

SM.3.3.21

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (2, -1, -1)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 1, 0, 0, 0)'$

SM.3.3.22

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 1, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.23

$$A = \begin{bmatrix} 0 & 2 & 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 1 & 2 & 1 & 1 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.24

$$A = \begin{bmatrix} 0 & 2 & 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (2, 1, -1)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 0, 0, 0, 0)'$

SM.3.3.25

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 2 \\ 2 & 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -0.5, -0.5)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 0, 1, 0, 0)'$

SM.3.3.26

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 2 & 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 0)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 1, 0, 0, 0, 1, 0, 0, 0)'$

SM.3.3.27

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 2 & 0 & 2 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.28

$$A = \begin{bmatrix} 0 & 2 & 1 & 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 2 & 0 & 2 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 1, 0, 0, 1, 0, 0)'$

SM.3.3.29

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 & 1 & 1 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 1)'; \alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 0, 0, 1, 0, 0, 0)'$

SM.3.3.30

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 1)'; \alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 1, 1, 0, 0, 0, 0)'$

SM.3.3.31

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-2, 1, 1)'; \alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 1, 0, 0, 0)'$

SM.3.3.32

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 1 & 0 & 0 & 2 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'; \alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 1, 0, 0, 0, 0)'$

SM.3.3.33

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'; \alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 1, 1, 1, 0, 0, 0)'$

SM.3.3.34

$$A = \begin{bmatrix} 0 & 2 & 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 2 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 1)'; \alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 1, 1, 1, 0, 0)'$

SM.3.3.35

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 0 & 0 & 2 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 2, 1)'$; $\alpha_t = (0, 1, 1)'$; $c = (1, 1, 1, 0, 0, 0)'$

SM.3.3.36

$$A = \begin{bmatrix} 2 & 0 & 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -2)'$; $\alpha_t = (-1, 1, 1)'$; $c = (1, 0, 1, 0, 0, 1, 0)'$

SM.3.3.37

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 1 & 1 & 2 \\ 2 & 2 & 2 & 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, -1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (0, 1, 0, 0, 1, 0, 0)'$

SM.3.3.38

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (2, -1, -1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (1, 0, 1, 0, 0, 0)'$

SM.3.3.39

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (0, 1, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.40

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, -1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (1, 0, 0, 0, 0)'$

SM.3.3.41

$$A = \begin{bmatrix} 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, -1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (1, 0, 0, 0, 0, 0)'$

SM.3.3.42

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, -1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.43

$$A = \begin{bmatrix} 2 & 0 & 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (1, 0, 1, 0, 0, 0, 1, 0)'$

SM.3.3.44

$$A = \begin{bmatrix} 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (2, 1, -1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (1, 0, 1, 0, 0, 0)'$

SM.3.3.45

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -0.5, -0.5)'$; $\alpha_t = (-1, 1, 1)'$; $c = (0, 1, 0, 0, 1, 0)'$

SM.3.3.46

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, -0.5)'$; $\alpha_t = (-1, 1, 1)'$; $c = (1, 0, 0, 0, 0, 0)'$

SM.3.3.47

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 \\ 2 & 2 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 0)'; \alpha_t = (-1, 1, 1)'$; $c = (0, 1, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.48

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'; \alpha_t = (-1, 1, 1)'$; $c = (0, 1, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.49

$$A = \begin{bmatrix} 1 & 0 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, 0)'; \alpha_t = (-1, 1, 1)'$; $c = (1, 0, 0, 0, 0, 0)'$

SM.3.3.50

$$A = \begin{bmatrix} 2 & 0 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 0)'; \alpha_t = (-1, 1, 1)'$; $c = (1, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.51

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'; \alpha_t = (-1, 1, 1)'$; $c = (1, 1, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.52

$$A = \begin{bmatrix} 2 & 0 & 1 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'; \alpha_t = (-1, 1, 1)'$; $c = (1, 0, 1, 0, 1, 0, 0, 1, 0)'$

SM.3.3.53

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 & 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 0 & 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.54

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (0, 1, 0, 1, 1, 0, 1, 0, 0)'$

SM.3.3.55

$$A = \begin{bmatrix} 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-2, 1, 1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (1, 0, 1, 0, 0, 0)'$

SM.3.3.56

$$A = \begin{bmatrix} 2 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (1, 0, 1, 0, 0, 0, 0)'$

SM.3.3.57

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 2 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (1, 1, 0, 1, 1, 0, 0, 0)'$

SM.3.3.58

$$A = \begin{bmatrix} 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 2 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 1)'$; $\alpha_t = (-1, 1, 1)'$; $c = (1, 0, 1, 1, 0, 1, 0)'$

SM.3.3.59

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 2, 1)'; \alpha_t = (-1, 1, 1)'$; $c = (1, 0, 1, 0, 0, 0)'$

SM.3.3.60

$$A = \begin{bmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -2)'; \alpha_t = (-2, 1, 1)'$; $c = (0, 1, 1, 0, 1, 0)'$

SM.3.3.61

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 0 & 2 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, -1)'; \alpha_t = (-2, 1, 1)'$; $c = (0, 1, 0, 1, 0, 0)'$

SM.3.3.62

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (2, -1, -1)'; \alpha_t = (-2, 1, 1)'$; $c = (0, 1, 1, 0, 0)'$

SM.3.3.63

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'; \alpha_t = (-2, 1, 1)'$; $c = (0, 1, 1, 0, 0, 1, 0, 0)'$

SM.3.3.64

$$A = \begin{bmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 0 & 1 & 2 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'; \alpha_t = (-2, 1, 1)'$; $c = (0, 1, 1, 0, 0, 1, 0)'$

SM.3.3.65

$$A = \begin{bmatrix} 0 & 2 & 1 & 2 & 2 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (2, 1, -1)'; \alpha_t = (-2, 1, 1)'$; $c = (0, 1, 1, 0, 0)'$

SM.3.3.66

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 2 \\ 0 & 2 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -0.5, -0.5)'; \alpha_t = (-2, 1, 1)'$; $c = (0, 1, 0, 1, 0)'$

SM.3.3.67

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'; \alpha_t = (-2, 1, 1)'$; $c = (0, 1, 1, 1, 0, 0, 1, 0, 0)'$

SM.3.3.68

$$A = \begin{bmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 0 & 1 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'; \alpha_t = (-2, 1, 1)'$; $c = (0, 1, 1, 1, 0, 1, 0)'$

SM.3.3.69

$$A = \begin{bmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 0 & 1 & 0 & 2 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'; \alpha_t = (-2, 1, 1)'$; $c = (0, 1, 1, 0, 0, 0)'$

SM.3.3.70

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'; \alpha_t = (-2, 1, 1)'$; $c = (0, 1, 1, 1, 1, 0, 0)'$

SM.3.3.71

$$A = \begin{bmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 0 & 1 & 2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 1)'$; $\alpha_t = (-2, 1, 1)'$; $c = (0, 1, 1, 1, 1, 0)'$

SM.3.3.72

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 2 \\ 0 & 1 & 0 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 2, 1)'$; $\alpha_t = (-2, 1, 1)'$; $c = (0, 1, 1, 0, 0)'$

SM.3.3.73

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, -1)'$; $\alpha_t = (1, 0, 1)'$; $c = (1, 1, 0, 0, 0, 0, 1, 0, 0)'$

SM.3.3.74

$$A = \begin{bmatrix} 1 & 2 & 1 & 1 & 2 & 1 & 2 & 0 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'$; $\alpha_t = (1, 0, 1)'$; $c = (1, 1, 1, 0, 0, 0, 0, 1, 0)'$

SM.3.3.75

$$A = \begin{bmatrix} 2 & 2 & 0 & 1 & 2 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 0)'$; $\alpha_t = (1, 0, 1)'$; $c = (1, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.76

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 & 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 0 & 1 & 2 & 2 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (1, 0, 1)'$; $c = (1, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0)'$

SM.3.3.77

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 & 2 & 2 & 0 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'; \alpha_t = (1, 0, 1)'$; $c = (1, 1, 1, 0, 1, 0, 0, 1, 0)'$

SM.3.3.78

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 2 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 1)'; \alpha_t = (1, 0, 1)'$; $c = (1, 1, 1, 1, 1, 1, 0, 0, 0)'$

SM.3.3.79

$$A = \begin{bmatrix} 2 & 0 & 1 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'; \alpha_t = (1, 0, 1)'$; $c = (1, 1, 0, 1, 0, 0, 1, 0, 0)'$

SM.3.3.80

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'; \alpha_t = (1, 0, 1)'$; $c = (1, 1, 1, 1, 0, 0, 1, 0, 0)'$

SM.3.3.81

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 & 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 2 & 1 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -2)'; \alpha_t = (0, 0, 1)'$; $c = (1, 1, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.82

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 2 & 1 & 0 & 2 \\ 0 & 2 & 0 & 2 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, -1)'; \alpha_t = (0, 0, 1)'$; $c = (1, 1, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.83

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 0 & 2 & 0 & 2 & 0 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'; \alpha_t = (0, 0, 1)'$; $c = (1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.84

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & 0 & 0 & 2 & 2 \\ 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, -1)'; \alpha_t = (0, 0, 1)'$; $c = (1, 0, 0, 0, 0, 0)'$

SM.3.3.85

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 1 & 1 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, -1)'; \alpha_t = (0, 0, 1)'$; $c = (1, 1, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.86

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, -1)'; \alpha_t = (0, 0, 1)'$; $c = (1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.87

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 & 1 & 0 & 2 & 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'; \alpha_t = (0, 0, 1)'$; $c = (1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.88

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 & 1 & 0 & 2 & 0 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 0 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, -0.5)'; \alpha_t = (0, 0, 1)'$; $c = (1, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.89

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 & 1 & 0 & 2 & 0 & 2 & 1 & 0 & 2 \\ 0 & 2 & 0 & 0 & 1 & 2 & 2 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 0)'; \alpha_t = (0, 0, 1)'$; $c = (1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.90

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 2 & 0 & 2 & 0 & 2 & 0 & 2 & 0 & 2 & 0 & 2 \\ 0 & 1 & 2 & 0 & 0 & 1 & 1 & 2 & 2 & 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'; \alpha_t = (0, 0, 1)'$; $c = (1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.91

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 2 & 2 & 0 & 0 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, 0)'; \alpha_t = (0, 0, 1)'$; $c = (1, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.92

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 1 & 0 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 0 & 0 & 1 & 2 & 2 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 0)'; \alpha_t = (0, 0, 1)'$; $c = (1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.93

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 0 & 0 & 2 & 2 & 2 & 0 & 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'; \alpha_t = (0, 0, 1)'$; $c = (1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.94

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 & 0 & 2 & 0 & 2 & 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 2 & 0 & 0 & 2 & 2 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'; \alpha_t = (0, 0, 1)'$; $c = (1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.95

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 & 0 & 2 & 1 & 0 & 2 \\ 0 & 2 & 1 & 1 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 1)'$; $\alpha_t = (0, 0, 1)'$; $c = (1, 1, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.96

$$A = \begin{bmatrix} 0 & 2 & 1 & 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 0 & 2 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'$; $\alpha_t = (0, 0, 1)'$; $c = (1, 1, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.97

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 & 2 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -2)'$; $\alpha_t = (-1, 0, 1)'$; $c = (0, 1, 1, 0, 1, 0, 0, 0)'$

SM.3.3.98

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 0 & 0 & 2 & 2 & 0 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, -1)'$; $\alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0, 1, 0, 0, 0, 0, 0)'$

SM.3.3.99

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'$; $\alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.100

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, -1)'$; $\alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0, 0, 0, 0)'$

SM.3.3.101

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 2 & 1 & 2 \\ 0 & 1 & 2 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, -1)'; \alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0, 0, 0, 0, 0)'$

SM.3.3.102

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 2 & 1 & 1 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, -1)'; \alpha_t = (-1, 0, 1)'$; $c = (0, 1, 1, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.103

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'; \alpha_t = (-1, 0, 1)'$; $c = (0, 1, 1, 0, 1, 0, 0, 0, 0, 0)'$

SM.3.3.104

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 2 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, -0.5)'; \alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.105

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 2 & 1 & 2 & 2 & 1 & 2 \\ 0 & 0 & 2 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 0)'; \alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.106

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 2 & 2 & 2 & 2 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'; \alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.107

$$A = \begin{bmatrix} 0 & 1 & 0 & 2 & 2 & 2 & 2 \\ 0 & 1 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, 0)'$; $\alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0, 0, 0, 0, 0)'$

SM.3.3.108

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 1 & 2 & 2 & 1 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 0)'$; $\alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.109

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 2 & 0 & 0 & 2 & 2 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (-1, 0, 1)'$; $c = (0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.110

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 0 & 1 & 2 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'$; $\alpha_t = (-1, 0, 1)'$; $c = (0, 1, 1, 0, 1, 0, 0, 0, 0)'$

SM.3.3.111

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 2 & 2 & 1 & 2 \\ 0 & 0 & 2 & 2 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 1)'$; $\alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0, 1, 0, 0, 0, 0)'$

SM.3.3.112

$$A = \begin{bmatrix} 0 & 2 & 0 & 1 & 1 & 2 & 1 & 2 \\ 0 & 1 & 2 & 0 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'$; $\alpha_t = (-1, 0, 1)'$; $c = (0, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.113

$$A = \begin{bmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 0 & 1 & 2 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'$; $\alpha_t = (-0.5, -0.5, 1)'$; $c = (0, 1, 1, 0, 0, 0)'$

SM.3.3.114

$$A = \begin{bmatrix} 0 & 2 & 1 & 2 & 2 \\ 0 & 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'$; $\alpha_t = (-0.5, -0.5, 1)'$; $c = (0, 1, 1, 0, 0)'$

SM.3.3.115

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, -1)'$; $\alpha_t = (1, -1, 1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.116

$$A = \begin{bmatrix} 2 & 1 & 1 & 0 & 2 & 2 & 0 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'$; $\alpha_t = (1, -1, 1)'$; $c = (0, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.117

$$A = \begin{bmatrix} 2 & 2 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 0)'$; $\alpha_t = (1, -1, 1)'$; $c = (0, 1, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.118

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (1, -1, 1)'$; $c = (0, 0, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.119

$$A = \begin{bmatrix} 2 & 2 & 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'; \alpha_t = (1, -1, 1)'$; $c = (0, 1, 1, 0, 1, 0, 0, 1, 0)'$

SM.3.3.120

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'; \alpha_t = (1, -1, 1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.121

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'; \alpha_t = (1, -1, 1)'$; $c = (0, 0, 1, 1, 1, 0, 1, 0, 0)'$

SM.3.3.122

$$A = \begin{bmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'; \alpha_t = (0, -1, 1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.123

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 0 & 2 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, -1)'; \alpha_t = (0, -1, 1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.124

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, -1)'; \alpha_t = (0, -1, 1)'$; $c = (0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.125

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 1 & 0 & 2 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'; \alpha_t = (0, -1, 1)'$; $c = (0, 0, 1, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.126

$$A = \begin{bmatrix} 0 & 2 & 1 & 1 & 0 & 2 & 1 & 0 & 2 \\ 0 & 0 & 2 & 1 & 2 & 2 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 0)'; \alpha_t = (0, -1, 1)'$; $c = (0, 0, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.127

$$A = \begin{bmatrix} 0 & 2 & 1 & 1 & 0 & 2 & 0 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & 2 & 1 & 1 & 2 & 2 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'; \alpha_t = (0, -1, 1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.128

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 1 & 0 & 2 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 2 & 2 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 0)'; \alpha_t = (0, -1, 1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.129

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'; \alpha_t = (0, -1, 1)'$; $c = (0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.130

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 1 & 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'; \alpha_t = (0, -1, 1)'$; $c = (0, 0, 1, 1, 1, 0, 0, 0, 0)'$

SM.3.3.131

$$A = \begin{bmatrix} 2 & 2 & 0 & 1 & 1 & 2 \\ 0 & 1 & 2 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -2)'$; $\alpha_t = (-1, -1, 1)'$; $c = (0, 1, 0, 1, 0, 0)'$

SM.3.3.132

$$A = \begin{bmatrix} 2 & 0 & 1 & 1 & 1 & 2 \\ 0 & 2 & 2 & 2 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, -1)'$; $\alpha_t = (-1, -1, 1)'$; $c = (0, 0, 1, 0, 0, 0)'$

SM.3.3.133

$$A = \begin{bmatrix} 2 & 0 & 1 & 0 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 2 & 2 & 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'$; $\alpha_t = (-1, -1, 1)'$; $c = (0, 0, 1, 0, 1, 0, 0, 0, 0)'$

SM.3.3.134

$$A = \begin{bmatrix} 2 & 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, -1)'$; $\alpha_t = (-1, -1, 1)'$; $c = (0, 1, 0, 0, 0)'$

SM.3.3.135

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, -1)'$; $\alpha_t = (-1, -1, 1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.136

$$A = \begin{bmatrix} 2 & 2 & 0 & 1 & 2 & 1 & 2 \\ 0 & 1 & 2 & 2 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'$; $\alpha_t = (-1, -1, 1)'$; $c = (0, 1, 0, 1, 0, 0, 0)'$

SM.3.3.137

$$A = \begin{bmatrix} 2 & 1 & 0 & 2 & 1 & 2 \\ 0 & 1 & 2 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, -0.5)'; \alpha_t = (-1, -1, 1)'$; $c = (0, 1, 0, 0, 0, 0)'$

SM.3.3.138

$$A = \begin{bmatrix} 2 & 0 & 1 & 1 & 2 & 1 & 2 \\ 0 & 2 & 2 & 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 0)'; \alpha_t = (-1, -1, 1)'$; $c = (0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.139

$$A = \begin{bmatrix} 2 & 0 & 1 & 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 1 & 2 & 2 & 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'; \alpha_t = (-1, -1, 1)'$; $c = (0, 0, 1, 0, 1, 0, 0, 0, 0)'$

SM.3.3.140

$$A = \begin{bmatrix} 2 & 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0.5, 0)'; \alpha_t = (-1, -1, 1)'$; $c = (0, 1, 0, 0, 0)'$

SM.3.3.141

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'; \alpha_t = (-1, -1, 1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.142

$$A = \begin{bmatrix} 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'; \alpha_t = (-1, -1, 1)'$; $c = (0, 1, 0, 1, 0, 0)'$

SM.3.3.143

$$A = \begin{bmatrix} 2 & 0 & 1 & 2 & 1 & 2 \\ 0 & 2 & 2 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 1)'$; $\alpha_t = (-1, -1, 1)'$; $c = (0, 0, 1, 0, 0, 0)'$

SM.3.3.144

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 & 2 & 2 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (1, -2, 1)'$; $c = (0, 1, 1, 1, 0, 0, 1, 0, 0)'$

SM.3.3.145

$$A = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 1)'$; $\alpha_t = (0.5, 0.5, 0.5)'$; $c = (1, 1, 1, 0)'$

SM.3.3.146

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 2 \\ 0 & 2 & 1 & 2 & 2 \\ 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, -1)'$; $\alpha_t = (0, 0.5, 0.5)'$; $c = (1, 0, 0, 0, 0)'$

SM.3.3.147

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'$; $\alpha_t = (0, 0.5, 0.5)'$; $c = (1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.148

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 & 0 & 2 \\ 0 & 1 & 1 & 2 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'$; $\alpha_t = (0, 0.5, 0.5)'$; $c = (1, 0, 0, 0, 0, 0)'$

SM.3.3.149

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 & 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 2 & 2 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'$; $\alpha_t = (0, 0.5, 0.5)'$; $c = (1, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.150

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 2 \\ 0 & 2 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, -1)'$; $\alpha_t = (-0.5, 0.5, 0.5)'$; $c = (1, 0, 0, 0, 0, 0)'$

SM.3.3.151

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'$; $\alpha_t = (-0.5, 0.5, 0.5)'$; $c = (1, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.152

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 & 0 & 2 \\ 0 & 2 & 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'$; $\alpha_t = (-0.5, 0.5, 0.5)'$; $c = (1, 0, 0, 0, 0, 0)'$

SM.3.3.153

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 2 & 2 & 0 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'$; $\alpha_t = (-0.5, 0.5, 0.5)'$; $c = (1, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.154

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 2 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, -1)'$; $\alpha_t = (-1, 0.5, 0.5)'$; $c = (0, 1, 0, 0, 0)'$

SM.3.3.155

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'$; $\alpha_t = (-1, 0.5, 0.5)'$; $c = (0, 1, 0, 0, 0, 0)'$

SM.3.3.156

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'$; $\alpha_t = (-1, 0.5, 0.5)'$; $c = (0, 1, 0, 0, 0)'$

SM.3.3.157

$$A = \begin{bmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'$; $\alpha_t = (-1, 0.5, 0.5)'$; $c = (0, 1, 0, 0, 0, 0)'$

SM.3.3.158

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 & 0 & 0 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (0.5, 0, 0.5)'$; $c = (1, 0, 0, 0, 0, 0, 0, 0, 0)'$

SM.3.3.159

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 2 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (0.5, -0.5, 0.5)'$; $c = (0, 0, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.160

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 & 0 & 2 & 1 & 0 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, 0)'$; $\alpha_t = (1, 1, 0)'$; $c = (1, 1, 0, 1, 1, 0, 1, 1, 0)'$

SM.3.3.161

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, -1, 1)'$; $\alpha_t = (1, 1, 0)'$; $c = (0, 0, 1, 1, 0, 0, 1, 0, 0)'$

SM.3.3.162

$$A = \begin{bmatrix} 1 & 1 & 2 & 2 & 2 & 0 & 1 & 1 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 1)'$; $\alpha_t = (1, 1, 0)'$; $c = (1, 0, 0, 1, 0, 1, 1, 0, 0)'$

SM.3.3.163

$$A = \begin{bmatrix} 2 & 2 & 2 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 0, 1)'$; $\alpha_t = (1, 1, 0)'$; $c = (0, 0, 1, 1, 0, 1, 0, 0, 0)'$

SM.3.3.164

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 0 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 2 & 0 & 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (1, 1, 0)'$; $c = (0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0)'$

SM.3.3.165

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 & 0 & 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 1)'$; $\alpha_t = (1, 1, 0)'$; $c = (1, 0, 0, 1, 1, 1, 1, 0, 0)'$

SM.3.3.166

$$A = \begin{bmatrix} 0 & 1 & 2 & 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 2 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'$; $\alpha_t = (1, 1, 0)'$; $c = (1, 0, 0, 1, 1, 0, 1, 0, 0)'$

SM.3.3.167

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 2 & 2 & 0 & 0 & 2 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'$; $\alpha_t = (1, 1, 0)'$; $c = (1, 0, 0, 1, 1, 1, 1, 0, 0)'$

SM.3.3.168

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 & 1 & 1 & 1 & 0 & 2 \\ 0 & 1 & 2 & 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, -1)'$; $\alpha_t = (0, 1, 0)'$; $c = (1, 0, 0, 0, 0, 1, 0, 0, 0)'$

SM.3.3.169

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 & 1 & 1 & 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 2 & 2 & 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'$; $\alpha_t = (0, 1, 0)'$; $c = (1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.170

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 & 1 & 1 & 0 & 2 \\ 0 & 2 & 2 & 1 & 1 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'$; $\alpha_t = (0, 1, 0)'$; $c = (1, 0, 0, 0, 0, 0, 1, 0, 0)'$

SM.3.3.171

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 & 1 & 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 0)'$; $\alpha_t = (0, 1, 0)'$; $c = (1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0)'$

SM.3.3.172

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 & 1 & 0 & 2 & 0 & 2 & 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 2 & 2 & 0 & 1 & 1 & 2 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'$; $\alpha_t = (0, 1, 0)'$; $c = (1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.173

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 2 & 0 & 1 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, -0.5, 0.5)'$; $\alpha_t = (0, 1, 0)'$; $c = (0, 0, 0, 1, 0, 0, 0, 0, 0)'$

SM.3.3.174

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & 2 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0.5, 0, 0.5)'$; $\alpha_t = (0, 1, 0)'$; $c = (0, 0, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.175

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 & 1 & 1 & 0 & 2 \\ 0 & 2 & 2 & 0 & 0 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -2, 1)'$; $\alpha_t = (0, 1, 0)'$; $c = (1, 0, 0, 1, 1, 0, 1, 0, 0)'$

SM.3.3.176

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 0 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, -1, 1)'$; $\alpha_t = (0, 1, 0)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.177

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 2 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, -1, 1)'$; $\alpha_t = (0, 1, 0)'$; $c = (0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.178

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 & 0 & 2 & 0 & 2 & 1 & 1 & 0 & 2 \\ 0 & 1 & 2 & 2 & 0 & 0 & 1 & 1 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 1)'$; $\alpha_t = (0, 1, 0)'$; $c = (1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0)'$

SM.3.3.179

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 0 & 2 & 1 & 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & 2 & 2 & 0 & 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 0, 1)'$; $\alpha_t = (0, 1, 0)'$; $c = (0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.180

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 & 2 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (0, 1, 0)'$; $c = (0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.181

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 & 0 & 2 & 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 2 & 2 & 0 & 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 1)'$; $\alpha_t = (0, 1, 0)'$; $c = (1, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0)'$

SM.3.3.182

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 2 & 0 & 0 & 2 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'$; $\alpha_t = (0, 1, 0)'$; $c = (0, 0, 0, 1, 1, 0, 0, 0, 0)'$

SM.3.3.183

$$A = \begin{bmatrix} 0 & 1 & 1 & 2 & 1 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, -1)'$; $\alpha_t = (-1, 1, 0)'$; $c = (0, 1, 0, 0, 0, 0, 1, 0, 0)'$

SM.3.3.184

$$A = \begin{bmatrix} 0 & 1 & 2 & 2 & 1 & 1 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'$; $\alpha_t = (-1, 1, 0)'$; $c = (0, 1, 0, 0, 0, 0, 0, 1, 0, 0)'$

SM.3.3.185

$$A = \begin{bmatrix} 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 0)'; \alpha_t = (-1, 1, 0)'$; $c = (0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.186

$$A = \begin{bmatrix} 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'; \alpha_t = (-1, 1, 0)'$; $c = (0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.187

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 0 & 2 & 2 & 0 & 0 & 1 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, -1, 1)'; \alpha_t = (-1, 1, 0)'$; $c = (0, 0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.188

$$A = \begin{bmatrix} 0 & 1 & 1 & 2 & 2 & 2 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 2 & 0 & 1 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 1)'; \alpha_t = (-1, 1, 0)'$; $c = (0, 1, 0, 0, 1, 0, 0, 1, 0, 0)'$

SM.3.3.189

$$A = \begin{bmatrix} 0 & 2 & 2 & 2 & 1 & 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 0, 1)'; \alpha_t = (-1, 1, 0)'$; $c = (0, 0, 0, 1, 0, 0, 0, 0, 0)'$

SM.3.3.190

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 & 0 & 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 2 & 2 & 0 & 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'; \alpha_t = (-1, 1, 0)'$; $c = (0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.191

$$A = \begin{bmatrix} 0 & 1 & 2 & 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 1)'$; $\alpha_t = (-1, 1, 0)'$; $c = (0, 1, 0, 0, 1, 0, 1, 0, 0)'$

SM.3.3.192

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (0.5, 0.5, 0)'$; $c = (0, 0, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.193

$$A = \begin{bmatrix} 2 & 0 & 1 & 2 & 2 & 2 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 & 1 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, -1)'$; $\alpha_t = (1, 0, 0)'$; $c = (0, 1, 0, 0, 0, 0, 1, 0, 0)'$

SM.3.3.194

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 & 1 & 1 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, -1)'$; $\alpha_t = (1, 0, 0)'$; $c = (0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0)'$

SM.3.3.195

$$A = \begin{bmatrix} 2 & 0 & 2 & 1 & 2 & 1 & 2 & 0 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 1, -1)'$; $\alpha_t = (1, 0, 0)'$; $c = (0, 1, 0, 0, 0, 0, 0, 1, 0)'$

SM.3.3.196

$$A = \begin{bmatrix} 2 & 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 0)'$; $\alpha_t = (1, 0, 0)'$; $c = (0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.197

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 1 & 2 & 0 & 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 & 0 & 0 & 1 & 2 & 2 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'; \alpha_t = (1, 0, 0)'$; $c = (0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0)'$

SM.3.3.198

$$A = \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 1 & 2 & 1 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-0.5, 0.5, 0.5)'; \alpha_t = (1, 0, 0)'$; $c = (0, 0, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.199

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 2 & 2 & 1 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0.5, 0.5)'; \alpha_t = (1, 0, 0)'$; $c = (0, 0, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.200

$$A = \begin{bmatrix} 2 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 2 \\ 0 & 2 & 0 & 0 & 1 & 2 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, -1, 1)'; \alpha_t = (1, 0, 0)'$; $c = (0, 0, 1, 0, 0, 1, 0, 0, 0)'$

SM.3.3.201

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 2 & 2 & 0 & 1 & 1 & 2 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, -1, 1)'; \alpha_t = (1, 0, 0)'$; $c = (0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.202

$$A = \begin{bmatrix} 2 & 1 & 2 & 0 & 2 & 0 & 2 & 1 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 1)'; \alpha_t = (1, 0, 0)'$; $c = (0, 0, 0, 1, 0, 1, 0, 0, 0)'$

SM.3.3.203

$$A = \begin{bmatrix} 2 & 2 & 2 & 0 & 1 & 0 & 1 & 0 & 1 & 2 & 2 & 2 \\ 0 & 1 & 2 & 0 & 0 & 1 & 1 & 2 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 0, 1)'; \alpha_t = (1, 0, 0)'$; $c = (0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0)'$

SM.3.3.204

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 0 & 0 & 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 0 & 1 & 2 & 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'; \alpha_t = (1, 0, 0)'$; $c = (0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.205

$$A = \begin{bmatrix} 2 & 0 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-2, 1, 1)'; \alpha_t = (1, 0, 0)'$; $c = (0, 1, 0, 1, 0, 1, 0, 1, 0)'$

SM.3.3.206

$$A = \begin{bmatrix} 2 & 0 & 1 & 2 & 0 & 1 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 & 0 & 0 & 2 & 2 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'; \alpha_t = (1, 0, 0)'$; $c = (0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0)'$

SM.3.3.207

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 0 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 & 2 & 0 & 2 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'; \alpha_t = (1, 0, 0)'$; $c = (0, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0, 0)'$

SM.3.3.208

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 & 2 & 0 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 & 1 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, -1)'; \alpha_t = (1, -1, 0)'$; $c = (0, 1, 0, 0, 0, 0, 1, 0, 0)'$

SM.3.3.209

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 1 & 2 & 2 & 1 & 1 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, -1)'$; $\alpha_t = (1, -1, 0)'$; $c = (0, 1, 0, 0, 0, 0, 0, 1, 0, 0)'$

SM.3.3.210

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 0)'$; $\alpha_t = (1, -1, 0)'$; $c = (0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.211

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (1, -1, 0)'$; $c = (0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.212

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 2 & 2 & 1 & 1 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, -1, 1)'$; $\alpha_t = (1, -1, 0)'$; $c = (0, 0, 0, 0, 0, 1, 0, 0, 0)'$

SM.3.3.213

$$A = \begin{bmatrix} 0 & 2 & 2 & 0 & 1 & 0 & 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 & 1 & 2 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 0, 1)'$; $\alpha_t = (1, -1, 0)'$; $c = (0, 0, 0, 1, 0, 1, 0, 0, 0, 0)'$

SM.3.3.214

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 2 & 0 & 0 & 0 & 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 2 & 2 & 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (1, -1, 0)'$; $c = (0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.215

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 & 0 & 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 & 2 & 2 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'; \alpha_t = (1, -1, 0)'$; $c = (0, 1, 0, 0, 1, 0, 0, 1, 0, 0)'$

SM.3.3.216

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 & 0 & 0 & 0 & 1 & 2 \\ 0 & 1 & 2 & 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'; \alpha_t = (1, -1, 0)'$; $c = (0, 1, 0, 0, 1, 0, 1, 0, 0)'$

SM.3.3.217

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'; \alpha_t = (0.5, 0.5, -0.5)'$; $c = (0, 0, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.218

$$A = \begin{bmatrix} 2 & 0 & 0 & 1 & 2 & 2 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'; \alpha_t = (1, 1, -1)'$; $c = (0, 0, 0, 1, 0, 0, 1, 0, 0)'$

SM.3.3.219

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 & 2 & 2 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'; \alpha_t = (1, 1, -1)'$; $c = (0, 0, 0, 1, 0, 0, 1, 0, 0)'$

SM.3.3.220

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 2 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, -1, 1)'; \alpha_t = (1, 1, -1)'$; $c = (0, 0, 0, 1, 1, 0, 0, 1, 0, 0)'$

SM.3.3.221

$$A = \begin{bmatrix} 2 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 0, 1)'; \alpha_t = (1, 1, -1)'$; $c = (0, 0, 0, 1, 1, 0, 1, 0, 0, 0)'$

SM.3.3.222

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'; \alpha_t = (1, 1, -1)'$; $c = (0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0)'$

SM.3.3.223

$$A = \begin{bmatrix} 2 & 0 & 0 & 2 & 0 & 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 1)'; \alpha_t = (1, 1, -1)'$; $c = (0, 0, 0, 1, 1, 1, 1, 0, 0)'$

SM.3.3.224

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 0 & 0 & 2 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'; \alpha_t = (1, 1, -1)'$; $c = (0, 0, 0, 1, 1, 1, 1, 0, 0)'$

SM.3.3.225

$$A = \begin{bmatrix} 0 & 2 & 1 & 1 & 1 & 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, -1)'; \alpha_t = (0, 1, -1)'$; $c = (0, 0, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.226

$$A = \begin{bmatrix} 0 & 2 & 1 & 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 0)'; \alpha_t = (0, 1, -1)'$; $c = (0, 0, 1, 0, 0, 0, 1, 0, 0, 0)'$

SM.3.3.227

$$A = \begin{bmatrix} 0 & 2 & 1 & 0 & 2 & 0 & 2 & 1 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 & 2 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'$; $\alpha_t = (0, 1, -1)'$; $c = (0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.228

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 0 & 2 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, -1, 1)'$; $\alpha_t = (0, 1, -1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.229

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, -1, 1)'$; $\alpha_t = (0, 1, -1)'$; $c = (0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.230

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 0 & 2 & 1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 1)'$; $\alpha_t = (0, 1, -1)'$; $c = (0, 0, 1, 1, 0, 0, 1, 0, 0, 0)'$

SM.3.3.231

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 1 & 1 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 0, 1)'$; $\alpha_t = (0, 1, -1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.232

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (0, 1, -1)'$; $c = (0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.233

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 1 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 1)'$; $\alpha_t = (0, 1, -1)'$; $c = (0, 0, 1, 1, 1, 0, 0, 0, 0)'$

SM.3.3.234

$$A = \begin{bmatrix} 2 & 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, -1, 0)'$; $\alpha_t = (-1, 1, -1)'$; $c = (0, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.235

$$A = \begin{bmatrix} 2 & 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (1, 0, 0)'$; $\alpha_t = (-1, 1, -1)'$; $c = (0, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.236

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, -1, 1)'$; $\alpha_t = (-1, 1, -1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.237

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (-1, 1, -1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.238

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 0 & 2 & 1 & 1 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, -1)'$; $\alpha_t = (1, 0, -1)'$; $c = (0, 0, 0, 0, 0, 0, 1, 0, 0)'$

SM.3.3.239

$$A = \begin{bmatrix} 0 & 0 & 2 & 0 & 1 & 2 & 2 & 0 & 1 & 2 \\ 0 & 2 & 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 0)'$; $\alpha_t = (1, 0, -1)'$; $c = (0, 0, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.240

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 & 0 & 1 & 2 & 1 & 2 & 0 & 1 & 2 \\ 0 & 2 & 0 & 0 & 1 & 2 & 2 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (1, 0, -1)'$; $c = (0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0)'$

SM.3.3.241

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 2 & 0 & 1 & 2 & 2 \\ 0 & 2 & 0 & 0 & 1 & 2 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, -1, 1)'$; $\alpha_t = (1, 0, -1)'$; $c = (0, 0, 1, 0, 0, 1, 0, 0, 0)'$

SM.3.3.242

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 2 & 0 & 1 & 2 & 1 & 2 \\ 0 & 2 & 0 & 1 & 1 & 2 & 0 & 0 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, -1, 1)'$; $\alpha_t = (1, 0, -1)'$; $c = (0, 0, 1, 0, 0, 1, 0, 0, 0, 0)'$

SM.3.3.243

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 2 & 2 & 2 \\ 0 & 1 & 2 & 0 & 0 & 1 & 1 & 2 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 0, 1)'$; $\alpha_t = (1, 0, -1)'$; $c = (0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0)'$

SM.3.3.244

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (1, 0, -1)'$; $c = (0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0)'$

SM.3.3.245

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 2 & 0 & 0 & 2 & 2 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 1)'$; $\alpha_t = (1, 0, -1)'$; $c = (0, 0, 1, 0, 1, 0, 0, 1, 0, 0)'$

SM.3.3.246

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 2 & 0 & 2 & 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 1)'$; $\alpha_t = (1, 0, -1)'$; $c = (0, 0, 1, 1, 0, 0, 1, 0, 0)'$

SM.3.3.247

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 2 & 0 & 1 & 1 & 2 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 1, 0)'$; $\alpha_t = (1, -1, -1)'$; $c = (0, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.248

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 2 & 0 & 0 & 1 & 2 \\ 2 & 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 1, 0)'$; $\alpha_t = (1, -1, -1)'$; $c = (0, 0, 1, 0, 0, 0, 1, 0, 0)'$

SM.3.3.249

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (-1, 0, 1)'$; $\alpha_t = (1, -1, -1)'$; $c = (0, 0, 1, 0, 1, 0, 0, 0, 0)'$

SM.3.3.250

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (1, -1, -1)'$; $c = (0, 0, 1, 1, 0, 0, 0, 0, 0)'$

SM.3.3.251

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

i) $(t', t) = (1, 0)$; $\alpha_{t'} = (0, 0, 1)'$; $\alpha_t = (1, 1, -2)'$; $c = (0, 1, 1, 1, 1, 0, 0, 0, 0)'$

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